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and Applied Sciences

Master's Thesis

Wearable IMU sensor-based motion tracking for post-stroke motor impairment assessment during activities of daily living

Author:

Tanguy Lewko

Supervisor(s):

Dr. Prabhat Pathak, Ph.D.

James Arnold, Ph.D. Candidate

Advisor(s):

Prof. Conor Walsh, Ph.D.

Prof. Auke Ijspeert, Ph.D.

Harvard Biodesign Lab

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Abstract

Post-stroke individuals face challenges incorporating their affected arm into activities of daily living (ADLs) due to impairments in motor function. Although task-oriented occupational therapy is a proven solution for enhancing functional skills, stroke survivors often struggle with motivation and lack feedback when attempting to increase the use of their affected arm during everyday activities. Continuous monitoring of ADLs is a viable solution to provide feedback on the quantity and quality of impaired arm functional movements and to promote adherence to home rehabilitation programs. Recent advances in machine learning and wearable inertial measurement units (IMUs) offer promising opportunities for remote monitoring of rehabilitation training. While the prior art has focused primarily on quantifying the usage of the affected arm, a comprehensive assessment of rehabilitation progress requires evaluating movement quality during functional tasks. The objective of this research is thus to develop a wearable IMU sensor platform to segment and evaluate stroke survivors' upper extremity motions during ADLs.

We leveraged two open-source IMU datasets to develop initial algorithms for assessing movement quality during ADLs. The first dataset contained healthy participants performing six unimanual upper extremity self-care ADLs and was used to develop a deep learning algorithm for precise time segmentation of ADLs. We developed a hybrid architecture combining convolutional and bidirectional long short-term memory layers to extract spatial and temporal information from the input IMU data. Our algorithm employs dense labeling for time-step resolution segmentation at a frequency of 33 hertz, while integrating past and future knowledge into current predictions through a context-aware framework. We achieved F1 scores of 0.83 and 0.87 in across-subjects and within-subjects evaluations, respectively. Additionally, we determined that segmentation accuracy was highest when using only the accelerometer and gyroscope data from two IMU sensors attached to the forearms. We also proposed a methodology for collecting an annotated naturalistic IMU-based ADL dataset in a controlled environment, using video ground truth to reliably label ADLs. To demonstrate the feasibility of our activity segmentation algorithm, we collected a dataset of 5 healthy participants and 5 post-stroke participants. We tested our algorithm on two mildly impaired post-stroke participants, highlighting the potential need for tailored models for post-stroke individuals due to the high variability of impairments within the population. The second open-source dataset was used to extract samples of healthy and post-stroke participants wearing IMUs on their upper body while drinking and brushing their teeth. We identified robust kinematic metrics that can be computed directly from accelerometer and gyroscope data and that differentiate between healthy and stroke participants. Finally, we developed a novel data-driven movement quality rating scale, the motion healthiness score (MHS), to incorporate identified metrics into a framework for naturalistic motion quality assessment. The MHS utilizes the Mahalanobis distance to measure the disparity between the movement of a stroke survivor and the same movement executed by healthy subjects. It showed desirable properties such as high consistency when rating sample motions from the same subject and good separability ($>80\%$ accuracy) between healthy and post-stroke individuals. The MHS offers an opportunity to assess impairment in a functional setting without evaluation by a therapist.

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Symbol table

Latex Symbol	Description
C	Confusion matrix $[-]$
SNR	Signal-to-noise ratio $[dB]$
SOR	Signal-to-offset ratio $[dB]$
P	Power (unit of the base signal)
α_s	Scaling factor $[-]$
α_z	Zoom factor $[-]$
R	Rotation matrix
K	Homogeneous rotation matrix
θ	Rotation angle $[rad]$
F_c	Compression factor $[-]$
M	Downsampling factor $[-]$
λ_S	SPARC $[-]$
$\mathbf{v}$	Angular velocity $[rad/s]$
w_c	Cutoff frequency $[hz]$
$\mathfrak{F}(\cdot)$	Fourier transform operator
F	F statistic $[-]$
η^2	Effect size $[-]$
p	p-value $[-]$
t	t-statistic $[-]$
Σ	Covariance matrix $[-]$
λ	Eigenvalue $[-]$
W	PCA projection matrix $[-]$
D_M	Mahalanobis distance $[-]$
D_B	Bhattacharyya distance $[-]$
MHS	Motion healthiness score $[-]$

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Chapter 1

Introduction

Stroke is a significant cause of long-term disability worldwide. According to the World Health Organization (WHO), 15 million people worldwide suffer from a stroke every year, with 5 million resulting in death and another 5 million causing permanent disability, placing a burden on individuals, families, and the healthcare system. Over 143 million years of healthy life are lost each year due to stroke-related death and disability. The lifetime risk of stroke has increased over the last 20 years by 50% and is now one in four people [1]. There are two main types of stroke: ischemic and hemorrhagic. Ischemic strokes are caused by an obstruction of blood flow to the brain. Due to the lack of blood flow, the brain cannot get enough oxygen and nutrients, and brain cells begin to die within minutes. Hemorrhagic strokes are caused by sudden bleeding in the brain. Leaked blood puts pressure on brain cells, damaging them. Both types can lead to neurological impairments that affect motor function, sensation, and cognition. Stroke survivors often suffer from physical disabilities like paralysis or weakness on one side of the body, reduced fine motor skills, and limited range of motion due to muscle stiffness [2]. After six months from a stroke, about 65% of patients are unable to use their affected hand for their usual activities [3].

Stroke rehabilitation aims to increase patients' functional independence at home and in the community [4]. The central nervous system has some plasticity and can re-organize its connections to recover degraded cognitive and motor functions. These changes can be favored by learning and practice [5]. Post-stroke progressive, skilled motor practice is critical for continued skill gains, as shown by neuroimaging studies [6, 7]. Task-oriented occupational therapy is thus a reliable and safe way to improve functional skills and quality of life in patients [8, 9]. Occupational therapists (OTs) work to facilitate and improve motor control in people who have had a stroke by making them practice activities of daily living (ADLs). These tasks include self-care tasks such as showering, dressing, toileting, and laundry as well as self-feeding tasks such as meal preparation, eating, and drinking.

From a functional perspective, there are three main consequences of stroke in the upper extremity (UE) [10]: learned nonuse, learned bad use, and forgetting. Initially, nonuse results from stroke-related impairments, but as time goes by, it can become a learned behavior. Learned nonuse refers to the acquired behavior of avoiding the use of the paretic arm during ADLs, even if it could be used, due to the development of a habit of not using it at all. Learned bad use describes the reinforcement of an abnormal motion strategy because it helps to perform tasks despite impairments. Finally, forgetting explains the fact that strokes can lead to transient retention of new skills despite extensive practice, causing even small practice breaks to lead to terrible rehabilitation results. A good functional rehabilitation program should aim to train the patient regularly while ensuring that they do not learn abnormal motion.

Although traditional home exercise programs developed by therapists can improve compliance, they often lack long-term monitoring and feedback, reducing motivation over time [11]. Recent advances in machine learning and wearable inertial measurement units (IMUs) provide opportunities for remote monitoring of home-based rehabilitation training, which can be a viable solution to provide feedback on impaired arm functional movement quantity and quality and foster adherence to home rehabilitation



programs. While prior work on wearable sensor-enabled interventions has focused mainly on quantifying the use of the affected arm [12, 13], a comprehensive assessment of rehabilitation progress requires evaluating movement quality during functional tasks. The objective of this research is thus to develop a wearable IMU sensor platform to segment and evaluate stroke survivors' upper extremity motions during daily activities. We decompose this objective into three research questions (RQ).

- RQ1** How can we reliably segment in time a subset of upper extremity ADLs using IMU sensor data?
- RQ2** Can we identify robust IMU-based kinematic metrics that differentiate between healthy and stroke participants during upper-extremity ADLs?
- RQ3** How can we incorporate identified metrics into a framework for naturalistic motion quality assessment?

In Chapter 2, we review the prior art that uses wearable IMU sensors to monitor arm movements after stroke. Then, in Chapter 3, we explain how we developed an upper-extremity ADL segmentation algorithm that utilizes IMU sensor data, using an open-source dataset of ten healthy subjects performing ADLs. Chapter 4 presents our protocol and methods for collecting a dataset of healthy and post-stroke participants performing ADLs. The participants perform naturalistic activities of daily living in a controlled environment. In Chapter 5, we demonstrate the proof-of-concept of our activity segmentation algorithm in two mildly impaired post-stroke participants. In Chapter 6, we leverage a second open-source dataset to determine a set of robust task-specific metrics that differentiate healthy and post-stroke motions during three component movements within common ADLs: brushing teeth, drinking transport, and drinking recovery. In Chapter 7, we establish a framework for computing the quality of motion of stroke patients during various activities of daily living. This framework calculates the motion healthiness score and measures the Mahalanobis distance to healthy motions.

Chapter 2

Prior art on wearable sensors for post-stroke upper extremity rehabilitation at home

Individuals post-stroke often experience motor impairments, and kinematic assessment is an objective method of quantifying and understanding these impairments. It helps occupational therapists establish customized rehabilitation programs, monitor progress, and develop new solutions due to a better understanding of post-stroke motion deficits. Healthcare professionals use standardized impairment measures to evaluate patients' functional ability and determine the most appropriate treatment plan. These include the Functional Independence Measure (FIM), the Modified Barthel Index (MBI), the Upper-Extremity Fugl-Meyer Assessment (UEFMA), the action research arm test (ARAT), the Wolf Motor Function Test (WMFT), and others. Among these, the UEFMA is one of the most commonly used assessments. It evaluates various aspects of motor function, including range of motion, coordination, and strength. Motion capture systems can provide precise kinematic assessments. In contrast to visual evaluation by a therapist, they enable objective numeric evaluations of movements and can complement traditional kinematic assessments.

Optoelectronic motion capture systems are widely regarded as the most accurate and reliable method for capturing human movement data. They use multiple cameras to localize reflective markers placed on the body, providing accurate 3D motion data for joints. These systems are reliable in very controlled environments, but they are high-cost and necessitate large space requirements and extended training and setup, so they cannot be used easily outside of the lab. Inertial measurement units (IMUs) are wearable sensors that combine measurements from accelerometers, gyroscopes, and sometimes magnetometers, allowing the measurement of accelerations, angular velocities, and sensor orientation. While kinematic joint angles and/or positions can be estimated from sensor fusion of the sensors within an IMU, there are complications such as drift that make IMUs less accurate than optical motion capture systems. A great advantage of IMUs compared to motion capture systems is that they are easy to set up and use outside the lab. All data are collected on the device and can be synced after hours of wearing the IMUs.

Kim et al. (2021) conducted a review on the use of wearable sensors for assessing and treating upper-extremity after stroke [14]. They emphasize that wearable sensors can broaden the scope of assessing UE movements beyond clinical settings. This could enable assessments to be carried out in home settings. Wearable sensors have created numerous possibilities for continuously obtaining measurements related to rehabilitation, rather than just at specific intervals with in-clinic assessments. Wang et al.[15] conducted a systematic review of wearable-sensing devices designed for upper-body rehabilitation. Accelerometers and IMUs were the predominant sensing technologies used, comprising 84% of the



articles reviewed. Common attachment points were the trunk, upper arm, forearm, wrist, and fingers. Despite the proliferation of wearable systems, the paper also highlights the scarcity of clinical evaluations performed. This points to the importance of more thorough research to provide evidence on the effectiveness of these technologies in real-world clinical settings.

Immediately following a stroke, patients tend not to use their affected upper limb in daily living, which can lead to learned non-use [10]. Initial nonuse occurs because of hemiparesis or sensory loss. When nonuse becomes habitual, even for activities the affected arm can do, nonuse can become a learned behavior. This phenomenon is detrimental to rehabilitation because it limits progress and could lead to forgetting already relearned abilities. Tracking and providing feedback on arm use quantity with wearable sensors can help reduce learned nonuse and has been the focus of numerous works. Most studies differentiate between functional and non-functional use of the arm. Functional use includes goal-directed motion, while non-functional use includes balancing the arms while walking or conversational gestures. David et al. (2021) [13] used two wrist-worn IMUs to compute a gross movement score to quantify arm use. They were able to detect functional activities with 50-60% accuracy and eliminate non-function activities with 90% accuracy. They showed that the arm use of the five stroke patients was biased towards the nonimpaired arm and that stroke participants had reduced total arm use compared to healthy controls. Bailey et al. (2014) used wrist-worn accelerometers to assess bilateral UE activity during ADLs and used two variables, the bilateral magnitude and the magnitude ratio, to distinguish between unilateral and bilateral UE tasks [16]. The general arm use computed from the accelerometer was found to have good psychometric properties and high correlation with FMA scores [17, 18, 19]. Zambrana et al. (2019) proposed a hierarchical approach using inertial sensors to first distinguish between arm movement and non-movements using a threshold-based algorithm and then between functional movements and non-functional movements using machine learning, in five ADLs and walking [20]. They achieved accuracies of 0.91 and 0.9 in intra and inter-subject testing. Lum et al. (2020) also showed the superiority of machine learning models compared to standard threshold-based algorithms because they are better at distinguishing functional and non-functional arm use [21].

In addition to using wearable sensors to assess paretic arm use, other studies have also explored the effects of newly developed algorithms coupled with regular feedback on arm usage given to stroke survivors wearing wearable sensors. The acceptability of such systems has also been investigated in a small population of post-stroke individuals. Rubio Ballester et al. (2015) [12] developed a low-cost wearable bracelet to motivate stroke participants to use the paretic arm more by providing feedback on the activity of the paretic arm. Giving feedback led to increased paretic arm activity, and post-test activity without feedback was higher than pre-test activity. Their study confirmed that monitoring the amount of arm use may increase the value of using the paretic limb and that repetitive exposure to reinforcement-based feedback may modify an individual's goal and self-representation. Chae et al. (2020) developed a home rehabilitation system to record the type and frequency of exercises performed by the user using a smartwatch and ML algorithms [22]. They found that their system was able to facilitate participation in in-home training and improved the functional scores measured with the Wolf Motor Function Test and the Upper Extremity Fugl-Meyer Assessment in 17 post-stroke subjects. Da Silva et al. (2018) conducted a study to analyze the activity responses of seven stroke patients who had recently suffered a stroke, within 28 days [23]. Patients were given a vibrating alert from a triaxial accelerometer wristband to prompt movement of the impaired arm if their hourly activity level decreased. The therapists and patients reviewed the movement activity data twice a week to determine a personalized prompt threshold. The mean activity increase following a prompt ranged from 11% to 29%. The authors also conducted a pilot randomized controlled trial [24]. In another study by Whitford et al. (2020), feedback only affected the perceived quantity of arm use and not actual arm use in eight post-stroke participants [25]. Although this is a less positive outcome, the authors state that perception of arm use can serve as a necessary first step in building awareness of paretic limb use and promoting the necessary behavioral changes to reduce nonuse. Lee et al. (2018) [26] developed an approach to detect both goal-directed movements during ADL and to determine appropriate feedback during two structured in-home rehabilitation exercises using two wearable sensors. They tested their algorithm on 20 stroke participants and proposed a feedback method to promote high-dose motor practice by



providing a reminder to try to use the affected arm at the time of a movement performed with the unaffected arm and based on relative arm usage throughout the day. In a survey, 91.7% of occupational therapists expressed willingness to use the system in their practice. Furthermore, 88.2% of stroke survivors indicated that they would use the system if recommended by their therapist. These results demonstrate the demand for such systems.

Parnandi et al. (2021) have developed a detailed way to quantify post-stroke UE rehabilitation training [27]. They use a deep learning algorithm, taking as input data IMU sensor data and counting the primitives of motion. Primitives of motion are the building blocks of functional upper extremity motions that have been developed by Schambra et al. (2019) [28]. Five primitives of motion were identified: reach, reposition, transport, stabilize, and idle.

Most of the literature on wearable IMU sensors used for home UE rehabilitation of post-stroke survivors focuses on measuring the extent to which the paretic arm is being used throughout the day, thereby focusing on nonuse. However, little research has been done on analyzing movement quality during ADLs, which is essential for a comprehensive assessment of rehabilitation training, also targeting bad use. This is a challenging task given the important role of contextual information on the activity being performed for giving a person relevant feedback [13]. In Chapter 3, we develop an algorithm to segment UE ADLs from IMU sensor data to enable such analysis.

Chapter 3

IMU-based segmentation of upper-extremity ADLs

During therapy, occupational therapists watch patients accomplish ADLs and assess their performance compared to what the motions would look like if they were to be done by a healthy person, using this analysis to suggest exercises targeting weaknesses. The knowledge of the task being done is essential for feedback and identifying abnormal behaviors. In this chapter, we develop an algorithm to segment UE ADLs using wearable IMU sensors in healthy participants.

Segmenting activities can offer numerous advantages to a motion monitoring system, encouraging stroke patients to train their paretic arm. First, it allows for targeted immediate feedback for recognized activities that can replace generic suggestions. For example, once we identify that patients are using their unaffected arm for an activity, we could encourage them to use their paretic arm. Moreover, it facilitates the computation of contextual, task-specific metrics, providing insights from qualitative and quantitative data. Once we know what activity is being performed, we can compute metrics and compare them against normative healthy data to characterize motor impairments. Detecting activities can enable consistent tracking of repeated activities throughout the days, offering valuable progression metrics for rehabilitation.

Activity recognition through wearable sensors for the upper body remains an underexplored paradigm. The complexity arises from the high variability in upper body motions, where individuals may perform the same activities differently. Most lower-body activities such as standing, sitting, walking, or running are repetitive and cyclical which enables clear definitions, but upper-body activities are harder to define. For example, people often reach from different initial positions to different final positions which affects how many different "reach" motions look. The beginning and end of a reaching motion can be difficult to identify as people may move their arms continuously. Additionally, data labeling for upper body motions is much more expensive as UE ADLs tend to be short. Finally, non-functional motions such as conversational and non-task-oriented gestures have a significant variance, making it more likely to misclassify motions.

We start by giving an overview of datasets and algorithms for IMU-based segmentation of upper-extremity ADLs in Section 3.1.

3.1 Dataset and literature review

3.1.1 Datasets in upper body ADL segmentation

There is a shortage of datasets whose data could be used to train models for recognizing upper body activities of daily living. The challenge arises from the need for standardized activity definitions, with



the same activities often described in multiple ways, and from the variability of selected activities. Additionally, labeling data has a high cost since people perform many different short-duration motions consecutively. The number of IMU sensors, placement, sampling rate, and other parameters also vary greatly between datasets.

Boukhenoufa et al. (2022) provided an overview of the latest advances in the use of wearable devices to collect data and machine learning algorithms to evaluate exercises [29]. This article provides valuable information on the main challenges faced by researchers in this field. The primary challenge is the limited quantity and quality of data, as the number of patients available for study is often restricted, and data collection can be a burden for them.

Finding open-source datasets well-suited for activity segmentation, especially tailored for stroke patients, poses an even more significant challenge. This is why, despite the ultimate goal being activity segmentation for stroke patients, the initial focus is on datasets involving healthy individuals. Additionally, there is a need to identify and categorize activities from the non-paretic arm of stroke patients, making it also worth detecting motions from an arm without impairment.

We now review existing open-source datasets for upper body activity segmentation and highlight their strengths and weaknesses.

Multi-Sensory Dataset [30]

Ruzzon et Al. (2020) introduced a multi-sensory dataset related to activities of daily living [30]. The dataset acquisition involved ten healthy volunteers, right-handed and aged between 22 and 28, who performed 180 ADL instances while wearing six 9-axis IMUs (two for each arm, one on the back, and one on the right thigh). The activities were labeled from videos recorded by an RGB camera, and the labels include both the activity name and the arm performing the featured uni-manual activities.

The featured activities are gathered in Table 3.1, as well as the event defining the start and the end of the activity:

Task	Start	End
Walk	First leg movement	Both legs stop
Sit Down	Adjust vertical position to sit	Sitting posture is relaxed
Stand Up	Shift body weight from chair to feet	Standing at normal height
Open Door	Hand touches the handle	Door is fully open
Close Door	Hand contact	Door is closed
Pour Water	Touching the bottle	Hand releases the bottle
Drink Glass	Hand contact on the glass	No contact with the glass
Brush Teeth	Toothbrush touches the teeth	Toothbrush leaves the mouth
Clean Table	Hand-table contact	Contact with the table ceases

Table 3.1: Activities featured in the Multi-Sensory dataset

Due to its detailed activity labels and the variety of activities being performed, this dataset holds excellent value for activity recognition. However, this dataset has weaknesses. It features a limited demographic representation with only ten young adult right-handed participants. This might limit the generalizability of the dataset to a broader population as the execution of daily activities may vary across age groups and handedness. It does not feature bi-manual tasks. The data is collected in a controlled environment, i.e., all participants perform activities in the same room setting, which might limit home-based activity representation.

StrokeRehab Dataset [31]

The StrokeRehab dataset [31] features 22 males and 28 females who have undergone stroke. 22 are right-side paretic, and 28 are left-side paretic. The age distribution of the patients has a mean of 58.4



(SD=13.0), and the mean time since the participants had a stroke is 5.4 years (SD=6.1). Participants took the Fugl-Meyer upper extremity assessment with a mean score of 42.8 (SD = 15.8). Six of them have a severe impairment level (FMA 0-25), 25 moderate (FMA 26-52), and 19 mild (FMA 53-65) [32]. The participants performed five times the following set of ADLs: brushing teeth, putting on deodorant, combing hair, drinking, and vertical and horizontal transports. Each ADL is performed with all its preparatory actions. For example, the drinking ADL consists of opening the bottle, pouring water into a glass, taking a sip, and closing the bottle. Twenty controls also performed the same tasks as the stroke participants. All participants wore a set of nine IMUs attached to the cervical vertebra C7, the thoracic vertebra T12, the pelvis, and both arms, forearms, and hands. The data also includes processed joint angles estimated from the IMUs. While this dataset is extremely valuable as it features stroke participants, its usability is diminished because of the absence of video or motion capture ground truth. Except for some very salient motions, isolating motions within an activity is challenging, which prevents us from using this dataset for activity recognition but makes it suitable for some qualitative analysis of stroke patient motion.

OPPORTUNITY Dataset [33]

The OPPORTUNITY dataset was devised to benchmark human activity recognition algorithms. Data is collected from daily living activities using a combination of body-worn sensors, object sensors, and ambient sensors in a smart-home environment.

Four participants perform 16 activities: opening and closing doors, the fridge, drawers, and the dishwasher, cleaning the table, drinking from a cup, and toggling a switch. While this dataset is used to benchmark many activity recognition algorithms, the featured actions make it less interesting than the Multi-Sensory Dataset as the greater part of the actions are similar, and the number of subjects is smaller.

PAMAP2 [34]

The PAMAP2 Physical Activity Monitoring dataset contains data on 18 activities performed by nine subjects wearing three inertial measurement units: one on the wrist of the dominant arm, one on the chest, and one on the dominant side's ankle. This dataset is also often used to benchmark activity recognition algorithms from IMU sensors. However, it mainly features lower extremity activities like walking, running, and cycling, and only a few manual activities (e.g., ironing, vacuum cleaning, and folding laundry). It has only one sensor for detecting arm motions.

Conclusion on datasets for upper body ADL segmentation.

Datasets suited for upper-extremity ADL segmentation from wearable sensors are not common and all substantially different, featuring different activities and sensor placements. Our review of the available datasets revealed that the most interesting dataset we can use to develop our algorithm is the Multi-Sensory Dataset, as it comprises a pertinent set of self-grooming and self-feeding activities performed with both hands, with 5 IMU sensors on the upper body (two per arm and one on the back) and adequate labeling of the ADLs being performed.

3.1.2 Algorithmic approaches for wearable IMU activity segmentation

Evaluating state-of-the-art approaches to upper body activity segmentation from wearable sensors presents a challenge due to the absence of a reference benchmark dataset encompassing a diverse range of upper extremity activities. Across various works, datasets vary in the number of sensors employed and their placement on different body parts. Furthermore, these studies assess different activities, and even when activities overlap between two datasets, their definitions may differ. For example, an activity labeled as eating could involve using a spoon, chopsticks, or hands. Additionally, some papers include a null category that contains every other activity that is not among the activities of interest.



In contrast, others focus on differentiating within classes of interest, leading to much higher reported accuracy.

This review goes beyond focusing solely on upper body activities, encompassing more work utilizing wearable IMU sensors for activity recognition. We separate this review into two categories of algorithms: deep learning algorithms and other algorithms, which typically rely on various machine learning algorithms and pattern recognition algorithms.

Non deep-learning algorithms

The following discusses non-deep learning approaches to recognizing human activities using wearable sensors, including methods such as pattern recognition and machine learning algorithms.

Lemmens et al. (2015) [35] used multi-array signal feature extraction and pattern recognition algorithms with 2D convolution to recognize upper extremity activities such as drinking, eating, and brushing hair from the 3D accelerometers, gyroscopes, and magnetometers attached on the dominant hand, wrist, upper arm, and chest of 30 healthy participants and one stroke patient. Their algorithms showed perfect accuracy for the stroke patient under standardized conditions. Still, they were much less effective under daily life conditions due to the high variation in how actions were performed, showing the limitations of such pattern recognition algorithms. Huynh et al. (2007) [36] investigated the use of different machine learning models, namely K-means, nearest neighbor, support vectors machine (SVM), and hidden Markov models (HMM), to detect both three high-level activities (preparing for work, going shopping, doing housework) and 16 low-level activities that are part of the high-level activities (e.g. drive car, eat, iron, brush teeth) from three accelerometers placed on the right wrist, the righthand side of the hip and the right thigh. They tested the different algorithms on a 10-hour naturalistic dataset, with slightly better accuracy than 80% for low-level activities and 92% for high-level activities.

Bao and Intille (2004) [37] collected user-annotated data from 20 subjects wearing five biaxial accelerometers. Several classifiers were tried using handcrafted features, such as mean energy, frequency-domain entropy, and correlation of acceleration data. The mean recognition accuracy was greater than 80% in 20 daily activities (both related to the upper body and the lower body). A study by Chen et al. (2021) [38] tested different machine learning algorithms (Decision Tree, SVM, XGBoost) to detect 19 different upper extremity ADLs. XGBoost gave the best results, giving a 90% accuracy on the seven 'easiest' tasks to detect and with no null class. Yang et al. (2008) [39] proposed a distributed recognition method to classify actions using a low-bandwidth wearable motion sensor network. Their main contribution was showing that the distribution of multiple action classes satisfies a mixture subspace model, with one subspace for each. Reyes-Ortiz (2013) [40] collected a dataset of static and dynamic lower-body ADLs carried with a waist-mounted smartphone with embedded inertial sensors. They used multiclass SVM, showing an accuracy of 96% over six lower-body ADLs.

Deep-learning algorithms

Although the first deep learning algorithm was first developed in the 1960s by Ivakhnenko and Lapa [41], the field has gained popularity only in the last decade, due to the increase of available computational resources through the development of Graphics Processing Units (GPUs) and Tensor Processing Units (TPUs), coupled with the development of techniques to train deep neural networks more effectively, like backpropagation [42]. Since the breakthrough made by AlexNet [43], deep learning has been proven effective in various fields, including computer vision, natural language processing, speech recognition, and time series forecasting. Of particular interest to us is its application in human activity recognition.

Achieving high performance in deep learning is a challenging and iterative process. It involves considering many combinations of parameters, while an exhaustive grid search across parameters is impractical due to limited computer resources. The design choices range from data input methodologies, such as window length, stride, context inclusion, and preprocessing, to broader architectural decisions involving



layer types, count, and width, as well as choices related to optimizers, learning rates, and the incorporation of data augmentation. These parameters are usually chosen by integrating knowledge specific to the field in the design choices and doing empirical validations on the dataset. The initial step towards developing accurate models is to synthesize existing knowledge. We therefore review the different methods used to segment upper body activities from wearable sensor data.

Various architectural choices are commonly used for Human Activity Recognition using wearable sensors. Among these, Convolutional Neural Networks (CNNs) are frequently employed. CNNs have demonstrated exceptional performance in various applications, including image classification, speech recognition, and natural language processing [44]. Recurrent neural networks have also shown encouraging results in various domains. These networks are specifically designed to consider the temporal dependencies of the data. They are commonly utilized in tasks like analyzing time series data for speech recognition and natural language processing. Hybrid models that leverage both CNN and RNN layers to integrate the strengths of each type have also demonstrated significant potential.

Hammerla et al. (2016) [45] extensively reviewed deep, convolutional, and recurrent models across three wearable sensor datasets. They point out that while convolutional neural networks work well with long and repetitive activities such as walking or running, recurrent neural networks should be favored for short-duration activities with a natural ordering as they benefit from contextualizing observation across long periods. Yin et al. (2022) [46] developed a 1-D convolutional neural network (CNN)-based bidirectional long-short-term memory (bi-LSTM) parallel model with attention mechanism, which outperformed CNN and RNN models in classification accuracy and computational time complexity on UCI (lower body activities: standing, sitting, laying, walking, downstairs, upstairs) and WISDM HAR (lower body activities: walking, upstairs, downstairs, sitting, standing, jogging) datasets. Baloch et al. (2022), introduced two sensor fusion strategies, early and late sensor fusion utilizing a CNN and a CNN-LSTM [47]. Their findings indicate that a late sensor fusion model based on CNN-LSTM enhances the recognition accuracy in similar activities. Panwar et al. (2019) [48] used a CNN architecture to classify three upper limb movements of the human arm, extension, flexion, and forearm rotation, to measure rehabilitation progress. The input data was the triaxial acceleration of the wrist. Their approach outperformed the classical ML algorithm on this task. In a recent study, Gomez-Arrunategui et al. (2022) used a single wrist-worn IMU to identify reaching movements in two datasets: a functional assessment dataset that involved functional assessment tasks, and an Activities-of-Daily-Living (ADL) dataset that included tasks like pizza-making and walking [49]. The study involved 12 stroke patients, and the researchers used a CNN to classify the different activities. The trained classifiers performed better when trained and evaluated using standardized functional assessment tasks. Additionally, the researchers noted that algorithms developed using data collected in clinical settings may not be as effective when applied to data collected in home environments. Kyritsis et al. (2020) [50] used a convolutional recurrent neural network to detect bites and used the bites distributions to determine the start and end of meals. David et al. (2023) [51] used an LSTM-RNN neural network to recognize six upper-extremity activities in stroke-affected participants by computing 20 handcrafted features. Kaku et al. (2022) [52] implemented a sequence-to-sequence model to predict the sequence of motion primitives from wearable sensor data. Zhu et al. (2018) [53] research aimed to detect high-level ADLs using temporal semantics and used a Sequence-to-Sequence model, too. High-level ADLs are complex activities consisting of temporal-dependent gestures, such as making coffee. Abdel-Salam et al. (2021) [54] reviewed top-performing techniques in human activity recognition based on wearable sensors. They concluded that hybrid approaches relying on both CNN and LSTM layers lead to the best performance.

Self-supervised learning has also shown promising results in Human Activity recognition due to its reduced dependence on labeled data. Zhang et al. (2023) reviewed the different self-supervised learning frameworks for time series analysis: generative-based, contrastive-based, and adversarial-based frameworks [55]. Taghanaki et al. (2021) [56] trained a neural network on raw accelerometer data to predict the value in the z-dimension using past value of this dimension as well as past and present motion in x and y dimensions. They then froze the model and transferred it to a supervised human activity recognition network. Their algorithm outperformed state-of-the-art in multiple datasets. Overall,



self-supervised learning is worth investigating when we have access to a large quantity of unlabeled data.

While most existing work focuses on classifying a pre-segmented window as part of an activity, little work has investigated the precise time segmentation of continuous, non-pre-segmented data. In the first case, the algorithm is expected to give a label to a whole window that contains only one activity; in the latter, the window might have multiple activities, and the algorithm is expected to determine the starting and ending points of the activities precisely.

This task has been studied slightly more in video data, in which it is referred to as Temporal Action Segmentation (TAS) [57].

3.2 Data preparation

To begin with, we synchronized the Multi-Sensory dataset by combining the data from each IMU sensor and aligning it with the labels. We have identified six UE ADLs that we want to segment as classes. These include Open Door, Close Door, Pour Water, Drink Glass, Brush Teeth, and Clean Table. Any other motion, such as Standing Up, Sitting Down, or Walking, is considered part of the negative class and we do not train the algorithm to detect those activities. Our algorithm uses the raw acceleration and gyroscope data as input, which was recorded at an IMU sampling frequency of 33 Hz.

Time series windowing for activity segmentation

The initial step in the activity recognition process is to segment the input signal. Human Activity Recognition (HAR) algorithms generally take a set of fixed-length segments, known as windows of signal, as input, and then associate one or more predicted labels with each window. These algorithms process each window separately in a sliding window framework. In this context, we explain the different informed choices that we make for our time series windowing framework.

The conventional method for labeling time series data for machine learning models and neural networks typically involves assigning a single label to each window. This approach is called sliding window labeling. The label is often determined by either the majority of timesteps within the window (see Figure 3.1), or the label of the last timestep. While adequate for specific tasks, such as continuous activities like walking or running, or scenarios where only the occurrence of a particular event matters, this approach proves less suitable for tasks involving the segmentation of fine motions. In such cases, the compromise on time resolution becomes a limitation.

To overcome this limitation, we have adopted a dense labeling framework [58]. In this framework, each time step is assigned a label, which the neural network must predict accurately. This approach enables precise segmentation at the timestep level. As shown in Figure 3.1, the comparison between sliding window labeling and dense labeling highlights the benefits of the latter. However, the cost of using dense labeling is an increase in complexity. The neural network is required to make many inferences to handle the finer granularity of the data.

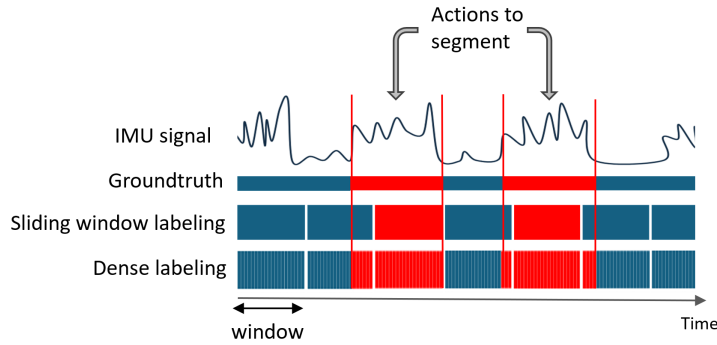


Figure 3.1: Sliding window labeling vs. dense labeling. Dense labeling is better for fine action segmentation due to its ability to contain multiple labels in each window, allowing for more granularity.

Two critical parameters frequently discussed are window length and stride, illustrated in Figure 3.2. Window length should ideally encompass a sufficient duration of motions to capture essential contextual information. For tasks involving fine-grained motions, it is desirable for the window to cover entire actions unless dealing with rhythmic actions like brushing teeth or cleaning a table, where its repetitive structure allows for information extraction without the necessity of the entire activity duration.

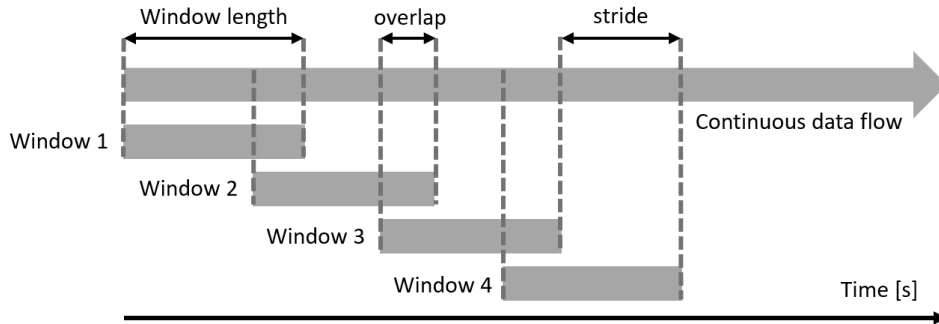


Figure 3.2: Windowing a time-series: window length, stride and overlap. Window length defines the size of each window, stride determines the step size between consecutive windows, and overlap specifies the amount of shared data points between adjacent windows.

Banos et al. (2014) [59] showed that large window sizes do not necessarily translate into better recognition performance and that optimal window length is activity-dependent. Jaén-Vargas et al. (2022) [60] showed that for the detection of walking, sitting and squatting, small windows (0.20-0.25s) showed similar performance as bigger ones (1-2s) while the inference time could be reduced by a factor 4 for LSTM and 2 for the CNN-LSTM model from windows of 100 frames (1s) to 20 frames (0.2s). While this result does not translate to upper-body activities, which include more semantics, especially for precise segmentation, it still points to the critical trade-off between window length and processing time. For upper-extremity activities, researchers tend to use longer windows of a few seconds to capture UE ADL patterns.

Established knowledge assumes that introducing overlap between consecutive windows helps prevent information loss at boundaries during training. However, Dehghani et al. (2019) [61] compared the use of non-overlapping windows and overlapping windows and showed that the performance improvements using overlapping windows reported in literature seem to be associated with the subject-dependent nature of the cross-validation process used. For subject-independent training, they concluded that when using subject-independent cross-validation, no improvement was found using overlapping windows.



We introduce a windowing framework designed for context-aware predictions. The objective is to avoid making predictions at window boundaries, where contextual information tends to be lacking. Leveraging both the left and right boundaries, representing past and future inputs, we enhance the model's capacity to predict labels at the center of the window, improving the predictive capabilities through a more comprehensive contextual understanding. Figure 3.3 illustrates the context-aware framework.

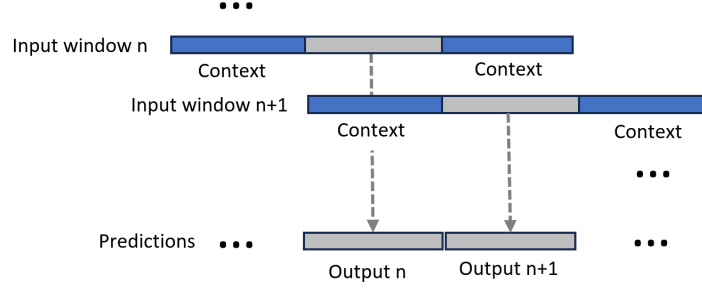


Figure 3.3: Context-aware prediction framework.

We use windows of 180 timesteps, corresponding to 5.45s, as it allows our algorithm to capture the temporal dependencies of upper extremity actions. We use the first 60 timesteps (1.82s) and the last 60 timesteps as context, and only the 60 middle timesteps are each paired with a corresponding label.

Managing imbalanced datasets

In multi-label classification scenarios, datasets often exhibit a substantial imbalance in the representation of each class label, i.e., specific labels occur much more frequently than others. Exposure to such imbalanced datasets can lead to model bias. Instead of discerning the distinctive features in the data that characterize the majority class, the model may somewhat default to predicting the majority class. This tendency arises because, in some cases, solely predicting the majority class and disregarding all other classes can yield better accuracy than trying to distinguish between all classes.

Standard solutions to the data imbalance challenge include random oversampling, undersampling, and weighted loss functions. Random undersampling removes instances from the majority classes until a more balanced distribution is achieved. This method was proven particularly effective in HAR with a large null class [44]. However, in some cases, particularly with smaller datasets, this method may result in a less accurate representation of the majority classes due to the potential loss of important information. On the contrary, random oversampling involves duplicating examples from minority classes. This strategy proves advantageous, especially for small datasets, as it avoids information loss. Nonetheless, a potential drawback is the risk of overfitting, given that exact replicas of examples are created. This could cause the model to memorize the examples instead of developing generalizable boundaries. Lastly, weighted loss functions can increase penalties for misclassification in the minority class. The weights can be inversely proportional to the frequency of classes. Doing so encourages the model to prioritize and improve its performance in underrepresented classes, fostering a more equitable learning process.

We opt for random oversampling due to the relatively small size of our dataset. We randomly sample the data with replacement until each class is represented by a minimum number of samples equal to the number of samples present in the majority class divided by 1.5, allowing us to balance all the classes we want to segment.

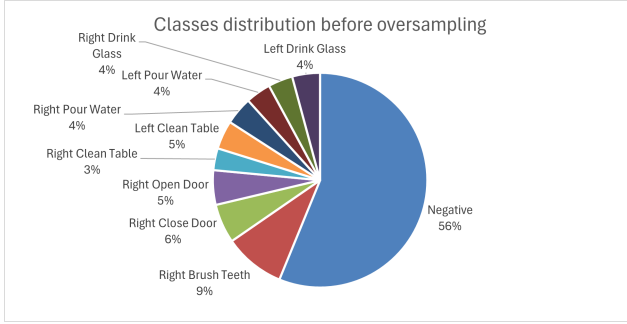


Figure 3.4: Classes distribution before oversampling.

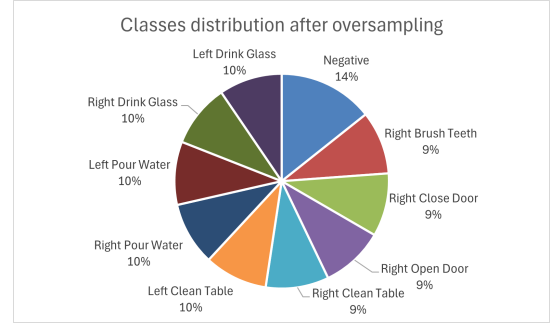


Figure 3.5: Classes distribution after oversampling.

3.3 Architectures and model selection

Obtaining an excellent performance in deep learning is an iterative and complex process. Indeed, there are a lot of parameters to take into account. There is a trade-off, as we cannot do an exhaustive grid search of parameters due to the computational resources in each model training, but we still want to reach the highest accuracy possible. We need to select the features to use as input to the model, the type of architectures and the natures of the layers being used, the number of layers and their width, the optimizer, learning rate, choice of data augmentation, etc. Often, we try to include as much knowledge from the field as possible to avoid trying many underperforming pipelines. We try to justify each choice either from literature or from experiments.

3.3.1 Validation strategies

Validation strategies are essential when assessing a model's performance, particularly in the context of upper body activity detection. The distinctive nature of datasets for such activities lies in the variability of how individuals perform the same activities. Subjects may exhibit variations in motion speed, employ different combinations of joint angles, and have slightly varying sensor placements and orientations. Two different types of models include personalized and generalized models. When aiming to assess the performance of a generalized model, testing should involve recordings from subjects not included in the training set. Conversely, training and testing datasets should encompass data from the same subjects for evaluating a personalized model.

Leave-One-Subject-Out (LOSO) validation is a form of cross-validation that involves training the model on all subjects but one and evaluating it on the remaining subject. This method assesses cross-subject generalization, uncovering potential biases or limitations specific to individual subjects [62]. If a generalized model works, user specific training data isn't needed, and we can use the same model universally.

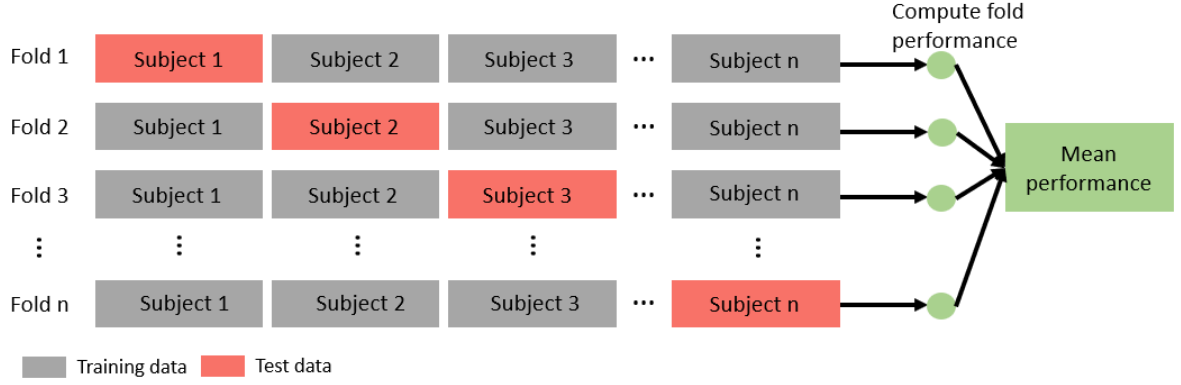


Figure 3.6: Leave-One-Subject-Out (LOSO) cross-validation process.

In within-subjects validation, each individual contributes to training and testing, capturing subject-specific patterns in both sets. Approximately 80% of the data from all subjects is utilized for training and model fitting, with the remaining 20% reserved for independent testing.

Employing both LOSO and within-subjects validation provides a nuanced assessment of the model's performance. LOSO enables the evaluation of a generalized model's performance. At the same time, within-subject validation offers more comprehensive insights into the model's performance, accounting for subject-specific nuances and allowing for rapid prototyping at a smaller computational cost.

3.3.2 Architecture

Given insights from the literature highlighting the effectiveness of LSTM and convolutional layers for activity recognition, we integrated these into our architecture.

The architecture presented in Figure 3.7 is a hybrid architecture that combines convolutional and recurrent layers to solve a multi-output classification task, predicting for each timestep both the activity being performed and which hand is performing. One example of output for a given timestep would be {brushing teeth, right}, meaning that the network predicts that the subject is brushing his teeth with his right hand.

The neural network takes an input window of the signal of size ws and with $\#features$ as the number of different input IMU channels. Then, a 1D convolutional layer is used to extract information on local patterns in the sensor data, followed by a Rectified Liner Unit (ReLU) activation to introduce non-linearity. Two bidirectional LSTM layers are then used to capture temporal patterns in the data sequences in the forward and backward directions. Finally, two fully connected layers are used for each classification task. The network outputs predictions only for the middle part of the signal to avoid making predictions on a signal that lacks context information.

We implemented multiple regularization techniques such as dropout, which is applied after each biLSTM layer, and L1 and L2 regularization, with value 10^{-5} for all model parameters.

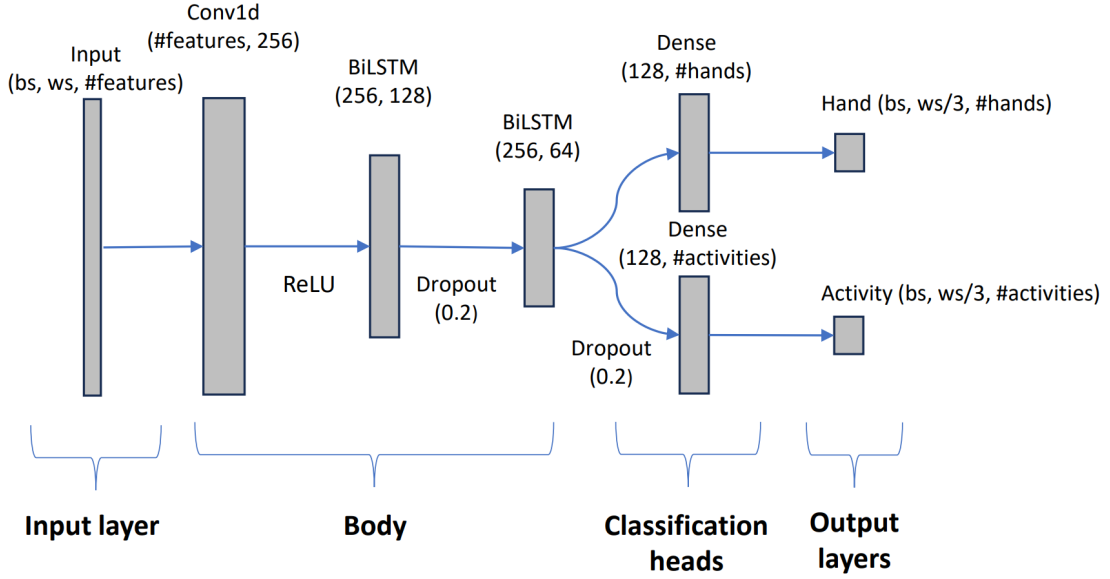


Figure 3.7: Hybrid architecture: Convolutional and BiLSTM layers

The loss is computed as the sum of the cross-entropy losses of the two classification tasks, and we chose Adam Optimizer [63] for gradient descent.

For comparison, we share the performance comparison with two other types of neural networks in Table 3.2, using only forearms IMUs accelerometer and gyroscope signals and within-subject evaluation. Reported metrics are explained in Appendix A. The LSTM architecture comprises three LSTM layers with decreasing sizes (256, 128, 64) and produces two sets of predictions with fully connected layers (output sizes 7 and 3). The biLSTM neural network is similar to the LSTM neural network, except all three LSTM layers are replaced with bidirectional LSTM layers. Using bidirectional LSTM layers led to a dramatic improvement in performance in the F1-score of 8.3% for the activity and 6.3% compared to regular LSTM layers. Replacing the first bi-LSTM layer of the biLSTM network by a 1D convolutional layer led only to a slight improvement (0.5% in activity and 0.7% in hand) but had the additional benefit of simplifying the model.

	Precision		Recall		Accuracy		F1-score	
	Act	Hand	Act	Hand	Act	Hand	Act	Hand
LSTM	0.766	0.811	0.776	0.771	0.766	0.811	0.764	0.811
biLSTM	0.846	0.874	0.848	0.874	0.846	0.874	0.847	0.874
CNN-biLSTM	0.852	0.880	0.854	0.880	0.852	0.880	0.852	0.881

Table 3.2: Performance comparison of LSTM, BiLSTM, and CNN-BiLSTM architectures

The implementation was done with Pytorch [64]. All training was done using a processor Intel(R) Core(TM) i7-8750H CPU @ 2.20GHz, 2208 MHz, six hearts, 12 logical processors, and an NVIDIA Quadro P1000 GPU (640 CUDA cores, 4 GB GDDR5 on-board memory, 128-bit memory interface, 82GB/s memory bandwidth).

3.3.3 Feature selection

Feature selection is crucial when developing a deep learning model. Excessive features can lead to data sparsity, especially in small datasets, potentially resulting in the model overfitting the data. Conversely,



too few features might make establishing boundaries between different instances challenging.

As we work with wearable sensors, the practicality and added burden of using more sensors is an important consideration [29]. The Multi-Sensory dataset comprises five sensors on the upper body: one on the back, one on each upper arm, and another on each lower arm. The minimum set of sensors to detect unimanual activities for both arms is one per arm, with the lower arm being intuitively preferable because of its proximity to the end-effector, i.e., there is greater signal change in a more distally placed IMU. Additionally, the placement of sensors on the lower arms aligns with the positioning of smartwatches, which would minimize the system's invasiveness, practicality, and cost.

Meng et al. (2021) [65] gathered evidence of sensor placement, giving more information. They investigated HAR performance involving stroke survivors using one single sensor from different body positions. The best accelerometer placements were brachioradialis and extensor carpi ulnaris (both localized on the forearm).

Table 3.3 compares CNN-biLSTM model performance on different sets of sensors. Interestingly, better performance was obtained with fewer features, with a 2.7% increase in within-subject F1-Score between a model with all five sensors and 50 features (x,y,z acceleration and gyroscope data and four quaternions) and a model taking only data from forearm IMUs and 12 features (acceleration and gyroscope data). Additionally, we investigated the effect of adding a 50% window overlap, which led to an improvement of 2.2% in the F1-score. It is essential to point out that, contrary to the increase in performance obtained by selecting fewer features, this performance increase is very costly in terms of computational resources, effectively doubling the number of windows we train on.

Performance is less good for LOSO cross-validation, with, respectively, 4.3% and 4.2% decrease in activity and hand F1 score. This can be explained by the fact that the training data does not include any subject-specific data used for validation. If one subject has a very different way of executing a motion or sensors are not placed in the same way, it will make it harder to detect by the algorithm that has not been trained on this kind of data.

Sensors	Channels	Number of features	Within-subjects weighted F1-score		LOSO weighted F1-score	
			activity	hand	activity	hand
	Acc(3), Gyr(3), Qat(4)	50	0.825	0.849	0.744	0.801
	Acc(3), Gyr(3)	30	0.848	0.871	N/a	N/a
	Acc(3), Gyr(3)	12	0.852	0.881	N/a	N/a
	Acc(3), Gyr(3) 50% window overlap	12	0.874	0.900	0.831	0.858

Table 3.3: Performance comparison with different sets of features/sensors.

In Figures 3.8 and 3.9, we show the confusion matrices of the best-performing model (CNN-BiLSTM, forearm IMUs, x, y, z acceleration, and gyroscope data) for within-subject validation. Performance varies depending on the activity. The best performance is obtained for brushing the teeth (precision = 0.98, recall = 0.98), probably because it is a very salient and specific motion for which the pattern is more straightforward to distinguish. The worst performance is for the pouring water activity (precision = 0.79, recall = 0.69), which is much shorter than brushing teeth and more complicated to identify.



There is very little confusion between drinking glass and brushing teeth, which both require hand-to-mouth motions, and we notice that, in general, the inter-class confusion remains very low, most of the errors involving the negative class. One exception is for pouring water, where 8% of the pouring water task is confounded with drink glass. We argue that the confusion with the negative class is less important than between the classes of interest because while they exclude some examples of motion from the analysis, they do not lead us to analyze some motions wrongly. Finally, the hand confusion matrix reveals a lower performance for left-hand attributed actions, reflecting that we have a lower amount of left-handed actions in the dataset: only cleaning the table, pouring water, and drinking from a glass were done with both right and left arm.

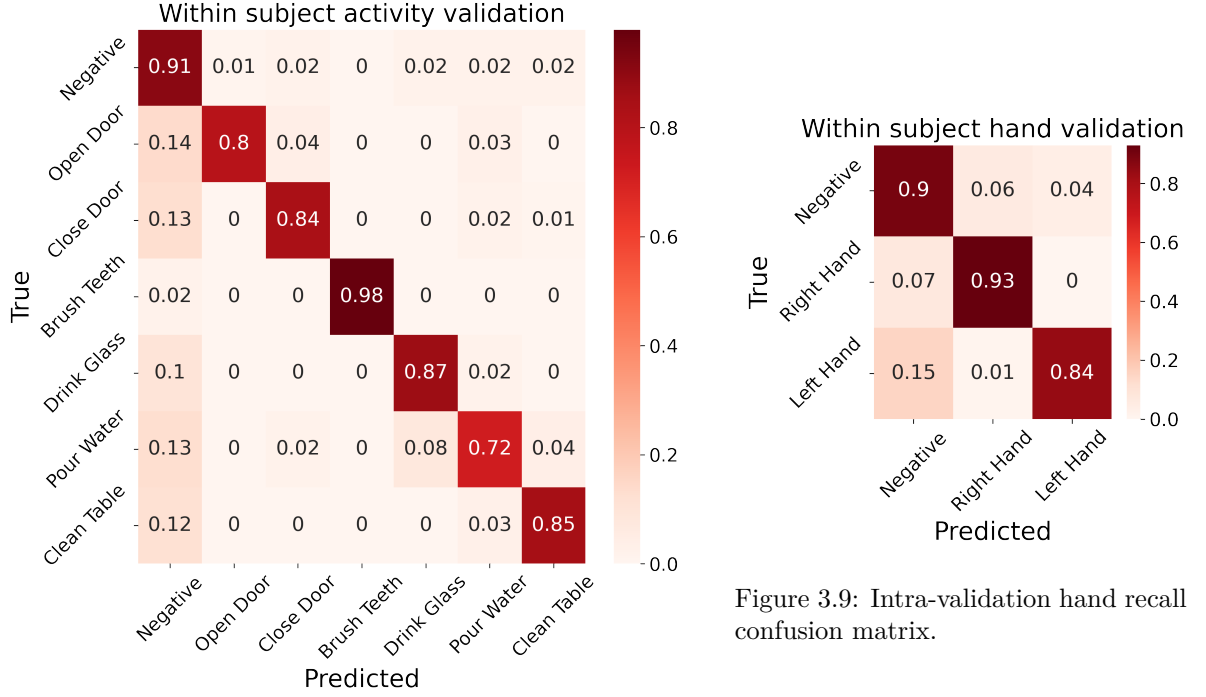


Figure 3.9: Intra-validation hand recall confusion matrix.

Figure 3.8: Intra-validation activity recall confusion matrix.

Table 3.4 shows the performance for each fold, i.e., left out subject, of the LOSO cross-validation. Performance varies significantly between the worst (activity F1-Score = 0.755) and the best predicted subject (activity F1-Score = 0.922). The mean of all subjects' activity F1-score is 0.831, with a standard deviation of 0.057. The hand prediction's mean is 0.857, and the standard deviation is 0.043.

Figures 3.10 and 3.11 show the confusion matrices for LOSO cross-validation with the same model. Compared to within-subject validation, some classes experience no or little difference in recall (Pour Water, Brush Teeth, Open Door, Negative) while others show a lower performance with LOSO validation (Clean Table -10%, Drink Glass -11%, and more importantly Close Door -22%).

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Activity F1	0.766	0.886	0.755	0.760	0.823	0.793	0.854	0.880	0.873	0.922
Hand F1	0.818	0.898	0.819	0.776	0.841	0.841	0.893	0.893	0.871	0.923

Table 3.4: LOSO cross-validation performance for each subject.

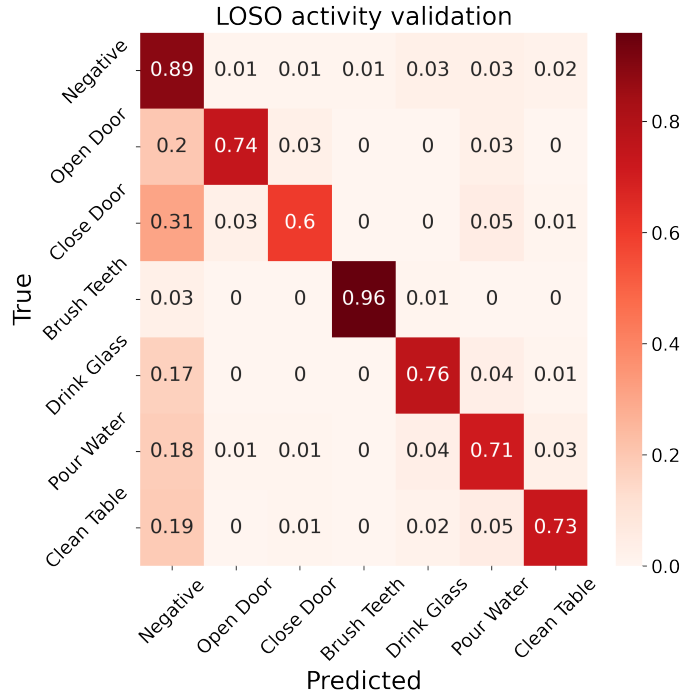


Figure 3.10: LOSO validation activity recall confusion matrix.

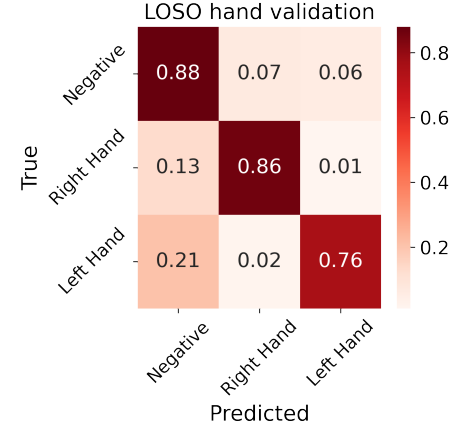


Figure 3.11: LOSO validation hand recall confusion matrix.

To better visualize results qualitatively, we show continuous activity segmentation visualizations. Figure 3.12 shows a close to average performance (85% accuracy). Figure 3.13 shows close to the best performance to expect (92% accuracy) with our current model. We show that for this period, the primary model error is predicting three times Close Door while the ground truth is Negative. We could not interpret false detections due to the absence of the video data shared in the dataset. A small fraction of the errors could be attributed to human error or ambiguity when labeling.

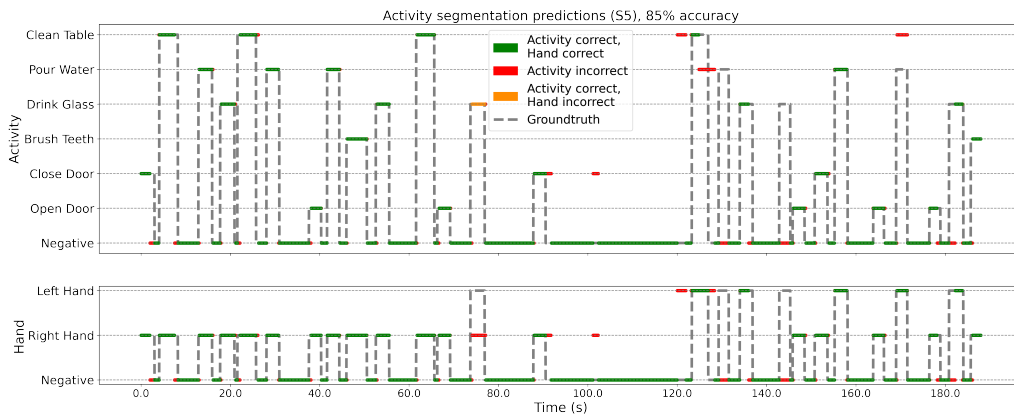


Figure 3.12: Example of close to average segmentation performance.

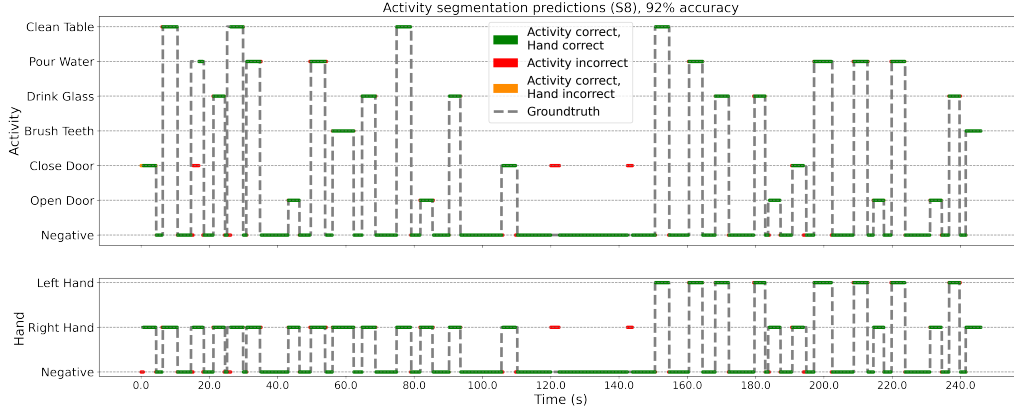


Figure 3.13: Example of close to best-case segmentation performance.

3.3.4 Data augmentations

When faced with a data shortage, data augmentations are an artificial solution to increase the size of the training set. The primary objective of data augmentation is to enhance the robustness of the trained model by introducing realistic transformations to the input data that mimic real-world scenarios. Wen et al. (2021) [66] review time series augmentations for deep learning and propose a taxonomy of time series augmentation techniques. We focus here on briefly exploring time-domain data augmentations.

Eyobu and Han (2018) applied data augmentations on spectral features data [67]. Um et al. (2017) [68] investigated data augmentation for wearable sensor data and their protocol to augment the data increased the classification performance of the motor state of Parkinson's disease patients from 77.54% to 86.88%. Uchitomi et al. (2023) also investigated data augmentations to classify Parkinson's disease and had the best results with magnitude-domain transformations such as jittering, rotating, and scaling compared to time-domain transformations such as time warping or cropping [69]. Their experiments with data augmentations included jittering, scaling, rotating, permuting, magnitude warping, and time warping. Rotation was identified as having the most significant positive effect on performance.

For time series data augmentation, several commonly utilized methods that we implemented include:

1. Jittering

Jittering aims to increase the robustness of the model to disturbances by adding noise to the data. When introducing jitter to a signal, it is essential to control the amplitude of the noise to prevent it from overshadowing the underlying signal. To do so, we define the maximum signal-to-noise ratio (SNR). The SNR is computed as the ratio of the signal power to the noise power and is expressed in decibels as follows:

$$SNR_{dB} = 10 \log_{10}(P_{signal}/P_{noise}) \quad (3.1)$$

Where P_{signal} is the signal power and P_{noise} is the noise power, the power is calculated as the mean value of the squared signal. Once we define a suitable SNR value, we can use it to compute the maximum noise power:

$$P_{noise} = \frac{P_{signal}}{10^{(SNR_{dB}/10)}} \quad (3.2)$$

We can then use this noise power to generate a random gaussian noise to add to the signal:



$$\text{Noise} \sim \mathcal{N}(0, \sqrt{P_{\text{noise}}}) \quad (3.3)$$

$$\text{Jittered signal} = \text{Signal} + \text{Noise} \quad (3.4)$$

2. Offsetting

The objective of offsetting is to increase the robustness of the model to constant bias by adding or removing constant value. Similarly to jittering, we compute the maximum offset value by setting a maximum Signal-to-Offset Ratio (SOR). However, a different random value was added to each timestep to jitter the signal. To offset the signal, we added the same value to all timesteps.

$$P_{\text{offset}} = \frac{P_{\text{signal}}}{10^{(\text{SOR}_{dB}/10)}} \quad (3.5)$$

$$\text{Offset Value} \sim \mathcal{U}\left(-\sqrt{P_{\text{offset}}}, \sqrt{P_{\text{offset}}}\right) \quad (3.6)$$

$$\text{Offsetted Signal} = \text{Signal} + \text{Offset Value} \quad (3.7)$$

3. Scaling

Scaling is used to make the model robust to variations of sensor sensitivity by multiplying the signal by a constant value. The effect of scaling is increased on peak values.

$$\text{Scaled Signal} = \alpha_s \times \text{Signal} \quad (3.8)$$

Where α_s is the scaling factor.

4. Zooming

Zooming mimics slowing down all motions by dilating the window in time. We zoom a signal by selecting a window and dividing the total number of timesteps in the window by a zoom factor (α_z), which gives us a reduced number of timesteps. We then select a corresponding continuous set of timesteps in the signal and scale it back to the original window size using linear interpolation.

5. Rotating

Rotating simulates minor differences in sensor positioning by rotating the sensors along one of its axes. Rotation matrices R_x , R_y and R_z for rotating by an angle θ the sensor signal are defined as:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix} \quad (3.9)$$

$$R_y(\theta) = \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \quad (3.10)$$

$$R_z(\theta) = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (3.11)$$



6. Time warping

Time warping is used to accelerate or slow down some parts of the window to make the model more robust to different speeds of motion. Our implementation of the time-warping data augmentation compresses 50% of the data chosen randomly by a compression factor F_c . It dilates the rest of the data so that the length of the signal stays the same. Compression and dilatation are done through linear interpolation.

7. Down-sampling

Down-sampling mimics a smaller frequency acquisition rate.

$$\text{Downsampled Signal} = \text{Signal}[k \cdot M] \quad (3.12)$$

where $k = 0, 1, 2, \dots, \lfloor \frac{ws-1}{M} \rfloor$ and M is the downsampling factor. ws is the window size.

Figure 3.14 illustrates the change in signal operated by some of the chosen data augmentation techniques, including jittering, offsetting, scaling, down-sampling, and time warping.

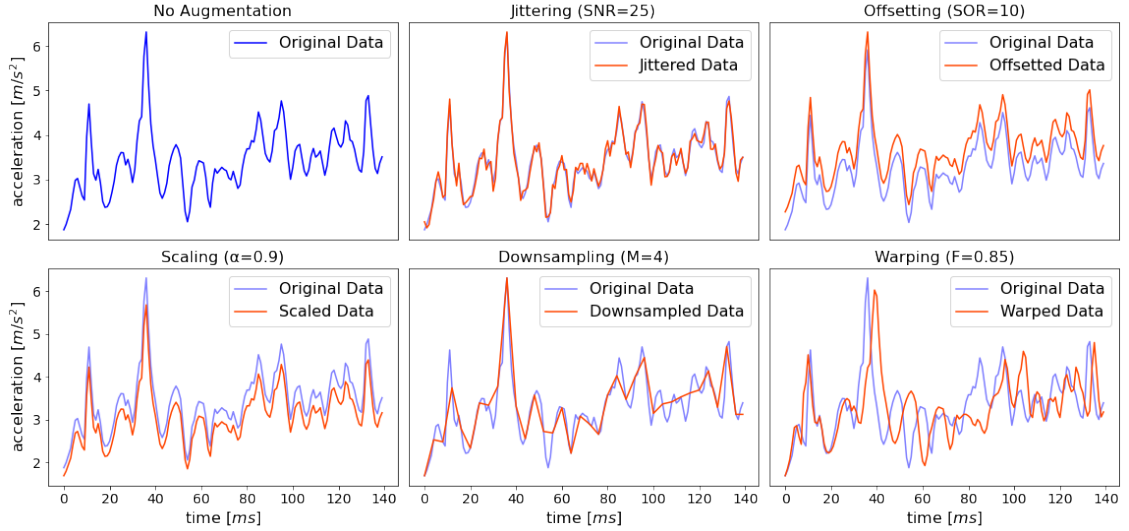


Figure 3.14: Data augmentations applied on an acceleration signal.

We used the following protocol to apply data augmentations when training our deep learning model. We assign probabilities to each augmentation to control how often it is applied.

Table 3.5 describes the probability and the parameters used for each applied data augmentation.

Augmentation	Parameter	Probability
Jittering	$\text{SNR} \in [25, \infty]$	0.3
Offsetting	$\text{SOR} \in [10, \infty]$	0.3
Scaling	$\alpha_s \in [0.9, 1.1]$	0.3
Zooming	$\alpha_z \in [1, 1.1]$	0.2
Rotating	$\theta \in [-15 \text{ deg}, 15 \text{ deg}]$	0.3 per IMU
Time warping	$F \in [0.85, 1]$	0.15
Downsampling	$M \in \{2, 3, 4\}$	0.15

Table 3.5: Data augmentation parameters and probabilities.



Training with data augmentations (see Table 3.6) led to improvements for all data augmentations, with the best improvement being for rotation (+1.4%).

	CNN-BiLSTM	Jitter	Scale	Rotation	Offset
F1-Score	0.852	0.853	0.861	0.868	0.859
	DownS	TimeW	Zoom		
F1-Score	0.854	0.856	0.853		

Table 3.6: Performance comparison with different data augmentations. Performance is evaluated with intra-subject validation, using forearm accelerations and angular velocities and a non-overlapping window.

More experiments should be conducted in the future to quantify the gain obtained from data augmentations in cross-subject validation and by combining multiple data augmentations in the same training.

3.4 Conclusion on upper-extremity ADL segmentation

Our activity segmentation pipeline is effective and appropriate for segmenting activities of healthy participants in active living settings. The boundaries of activities are segmented with great precision and we achieved an F1-score of 0.87 in within-subjects validation and 0.83 in across-subjects validation. The 4% decrease in performance when using LOSO validation indicates the importance of incorporating subject-specific knowledge into the training data. Additionally, our experiments were conducted in a controlled environment, and it is crucial to explore how this algorithm can generalize to the home environment. Finally, it is important to investigate how our algorithm would generalize to stroke patients due to the variability in their motions caused by impairment. To do so, we collect our own dataset, with a protocol described in Chapter 4. In Chapter 5, we use our algorithm to segment activities in two mild stroke participants.

This algorithm can be used to quantify the usage of the arms during specific ADLs and enables further kinematic analysis on pre-segmented ADLs. In Chapter 6, we examine differences in the execution of specific upper-extremity functional activities of daily living (ADLs) between healthy individuals and stroke patients.

Chapter 4

Collecting an annotated naturalistic dataset of post-stroke ADLs in a controlled environment

In this Chapter, we present the protocol used for building a dataset consisting of stroke participants performing upper-extremity ADLs. This dataset addresses the limitations that open-source datasets have. It provides synced video data and ground-truth kinematics from optical motion capture.

4.1 Protocol for unstructured ADL data collection

We use the Movella Xsens DOT [70] wearable sensor development platform, which features sensors that incorporate a 3D accelerometer, gyroscope, and magnetometer. The sensors work wirelessly, making them suitable for our unstructured ADLs experiment. We use a set of 10 sensors placed on the forehead, forearms, upper arms, hands, lower back, upper back, and sternum as depicted in Figure 4.1.

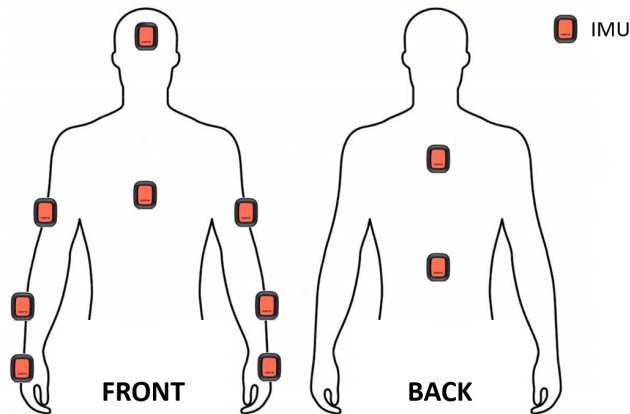


Figure 4.1: IMU placement on the upper body

The experiment is recorded using an HD camera that films at 60fps. In addition, Qualysis motion capture cameras are utilized to capture motion.



The signals are acquired and synced using a data acquisition system (Powerlab, ADInstruments) and recorded with a sampling rate of 200Hz per channel using appropriate software (Labchart, ADInstruments). Participants are screened before the experiment and their fine motor control and mobility levels are assessed to qualify them for the protocol.

After taking the Fugl-Meyer upper extremity assessment, the participants perform a circuit of ADLs consisting of washing their face, brushing their teeth, combing hair, applying deodorant, folding a towel, stirring a pot, pulling a chair to sit, drinking from a bottle, eating with a spoon from a bowl, and donning a hat. The experimental setup is represented in Figure 4.2, and we illustrate some ADLs in Figure 4.3. We selected these ADLs because they represent self-feeding and self-grooming tasks, essential for independence and improvement of quality of life, and reduced caregiver burden.

The circuit is done seven times for each participant under three conditions: A, B, and C. In condition A, which is repeated three times, participants are asked to perform the task in the same way they would perform it at home, as naturally as possible. In condition B, performed three times, an occupational therapist encourages using their impaired (non-dominant for healthy) arm. Finally, in condition C, participants wear a constraint-induced movement therapy glove (CIMT) (see Figure 4.3) to limit as much as possible the use of their non-impaired / dominant arm.

Condition A allows us to understand the participants' natural way of performing ADLs and quantify the use of their impaired arm across the various activities. It serves as a baseline. Condition B evaluates the impact of occupational therapist guidance on impaired arm use. Finally, condition C assesses the feasibility of each activity using only the impaired arm. These conditions allow us to obtain data for ADLs performed with both arms.

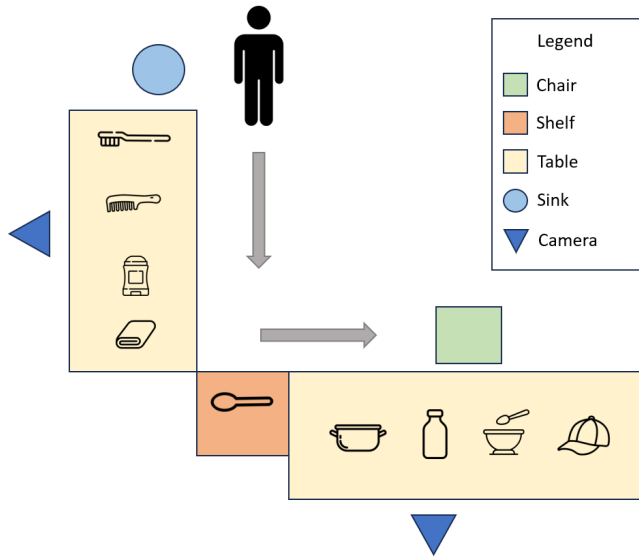


Figure 4.2: 11-ADL circuit set up.

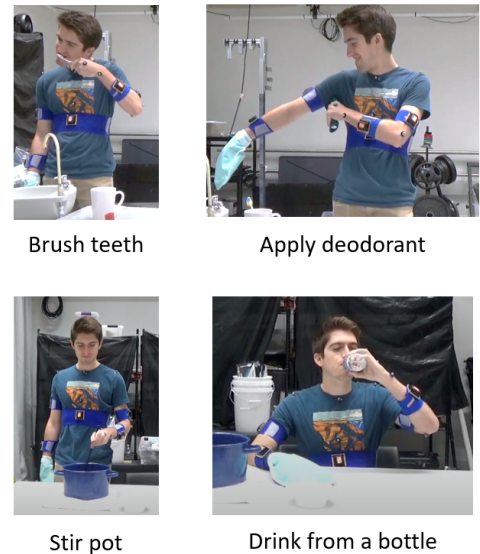


Figure 4.3: Examples of ADLs (CIMT glove on).

4.2 Labeling methodology

To analyze the data of the unstructured ADL protocol, we label upper extremity ADL segments that have fixed boundaries, are required to complete activities, and can be potentially valuable for analysis. A pertinent instance is the motion involved in tooth brushing, which begins with brush contact with teeth and ends with its disengagement. Reaching motions are omitted from labelization due to their considerable variability, identification challenges, and lack of clear definition.



Each categorized movement is linked to the start and end times, marked by discrete events. The list of 25 labeled ADL segments, together with their respective initiating and concluding events, is presented in Table 4.1. We also indicate the hand performing the task.

Activity	Start Time	End Time
Wash face - Hand to face	Hand starts moving upward	Contact with face
Wash face - Rubbing	Contact with face	Face contact ends
Wash face - Turn handle	Hand touches handle	Hand leaves handle
Brush teeth - Apply toothpaste	Tube touches brush	Tube leaves brush
Brush teeth - Hand to face	Toothbrush moves upwards	Contact with mouth
Brush teeth - Brushing	Toothbrush touches teeth	Toothbrush leaves mouth
Brush teeth - Recovery	Toothbrush leaves mouth	Toothbrush touches table/cup
Brush teeth - Rotate cap	Cap rotation starts	Cap on/removed
Comb - Hand to head	Comb leaves table	Comb touches head
Comb - Comb	Comb in contact with head	Last contact of comb with head
Comb - Recovery	Last contact of comb with head	Comb touches table
Apply deodorant - Rubbing	Deodorant touches armpit	Deodorant leaves armpit
Apply deodorant - Recovery	Deodorant leaves armpit	Deodorant touches table
Laundry - Fold	Start first fold	End last fold
Laundry - Unfold	Start unfolding	End unfolding
Pull Chair - Pull	Chair moves	Chair stops moving
Stirring pot - Stirring	Spoon touches pot	Stirring stops
Drinking - Rotate cap	Cap rotation starts	Cap on/removed
Drinking - Hand to face	Bottle leaves table	Contact with mouth
Drinking - Drinking	Bottle touches mouth	Bottle leaves mouth
Drinking - Recovery	Bottle leaves mouth	Bottle touches table
Eating - Hand to face	Spoon leaves plate	Spoon in the mouth
Eating - Recovery	Spoon leaves mouth	Spoon touches plate
Wear hat - Hand to head	Hat leaves table	Hat on head
Wear hat - Recovery	Hat leaves head	Hat touches table

Table 4.1: Definitions of activity labels

We labeled the activities using an open-source Python-based labeling software [71], built with QT, a cross-platform software for creating applications. Minor modifications were applied to the source code to suit our application.

The GUI allows one to open a video, visualize it, and move it frame by frame, select frames as the starting and ending times of an activity, select the activity, and add it. Then, the labels can be exported as a CSV table. The GUI also supports opening a CSV table to continue previously labeled videos. Labeling the data from one participant doing the unstructured ADL protocol with the labels defined in Table 4.1 takes about 3.5 hours and results in approximately fifty segmented actions.

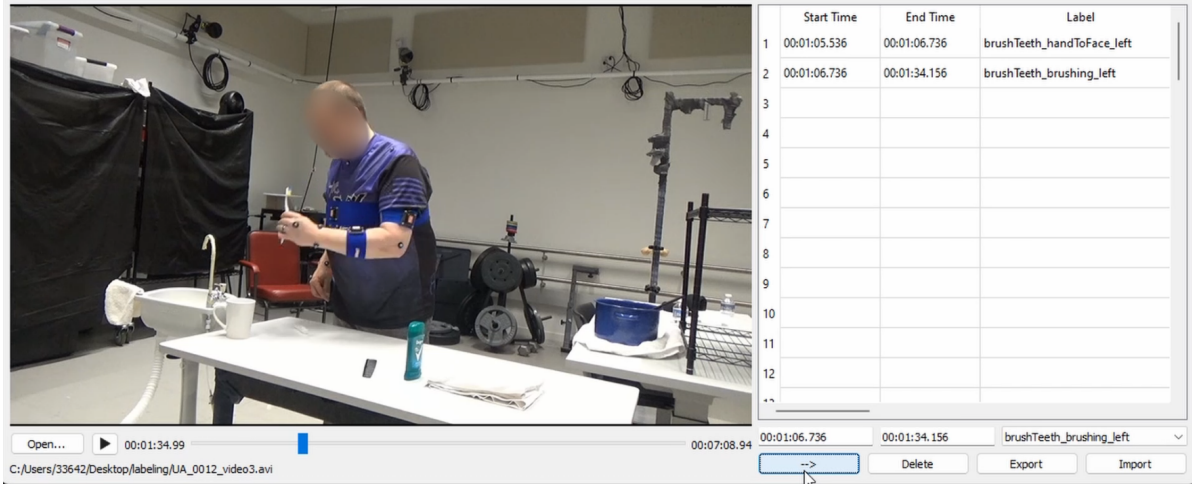


Figure 4.4: Labeling GUI. Adapted from [71].

4.3 Intermediary dataset characterization

We state the condition of the dataset as of the date this thesis was written. Table 4.2 provides information about the participants who took part in the study.

Participant	Sex	Age	Impaired arm	UE-FMA	Time since stroke
GG87	M	41	left	31/60	3 years 6 months
GG95	F	62	right	37/60	7 years
GG98	M	49	left	55/60	3 years 10 months
GG100	M	64	left	20/60	13 years
GG102	M	74	left	53/60	4 years 6 months
UA016	M	25	-	-	-
UA020	F	26	-	-	-
UA021	M	32	-	-	-
UA022	F	23	-	-	-
UA025	M	23	-	-	-

Table 4.2: Participant data collection summary

For each ADL run of the dataset, we provide videos (raw and blurred), optical motion capture data files, video-based labels, and IMU data (raw and synced with labels). On average, every run takes around 3.5 to 7 minutes. However, this duration may vary based on the speed at which the participants complete the tasks. In each run, there are approximately 50 actions that need to be identified by determining their start and end frames following the guidelines in Table 4.1.

This dataset was collected close to the end of this thesis, with much time dedicated to data acquisition, labeling, syncing, and blurring the videos. As a consequence, only a portion of the dataset was prepared for analysis at the time of the writing. Nonetheless, we were able to construct a proof-of-concept and some initial testing using the labeled data from stroke participants GG98 and GG102, and healthy participants UA016, UA020, UA021, and UA025.

Chapter 5

Proof-of-concept on post-stroke participants

In this Chapter, we present a proof-of-concept of our activity segmentation algorithm on stroke participants GG98 and GG102 from our collected data. We use our activity segmentation algorithm to segment new activities for the participants. We also show that we can decompose segmented activities even more into analyzable motions using the same algorithm.

5.1 Two-stage activity segmentation

We test our upper extremity activity segmentation algorithm developed in Chapter 3 on our dataset (see Chapter 4) to verify that our algorithm can effectively be transferred to new tasks performed by stroke participants. We tested our algorithm on two mildly impaired stroke participants, GG98 (UEFMA=55/60) and GG102 (UEFMA=53/60). The assessment revealed that both GG102 and GG98 lack shoulder abduction and flexion and present diminished coordination and speed. Additionally, GG98 lacks wrist circumduction and GG102 presents some flexor and extensor synergies.

We trained two different models. The first one aims to detect general activities. We selected a subset of six unimanual tasks (see Table 4.1 for definitions of the tasks): Brush teeth - Brushing, Comb - Comb, Apply deodorant - Rubbing, Stirring pot - Stirring, Eating (from first Hand-to-Face to last Recovery), and Drinking (from the start of Hand-To-Face to the end of Recovery). All other motions are part of the negative class. The second model aims to segment specific motions within a meal, i.e., within a combination of eating and drinking events. We segment the transport (lifting the spoon to the mouth) and recovery (lowering the spoon) phases of eating as well as the transport (lifting the bottle to the mouth), ingestion (drinking water), and recovery phases of drinking (lowering the bottle from the mouth to the table). Running the second model results in directly analyzable phases of an activity. The second model can be applied after the first one outputs the eating and drinking activity segments as shown in Figure 5.1.

The Multi-Sensory dataset was used to develop our activity segmentation algorithms and had about 15 repetitions of each task for ten subjects. Compared to the Multi-Sensory dataset, our dataset was much smaller when this thesis was written. We only had access to data from four healthy subjects and two post-stroke participants. Our dataset consisted of about 2 to 4 samples of the same activity per hand for each subject. This amount of data is minimal for training a deep-learning algorithm and the following results should thus be interpreted cautiously.

We conducted two different model training settings. In the first setting, we trained a *generalized* model to determine if a model trained on data from four healthy participants could be applied to each stroke

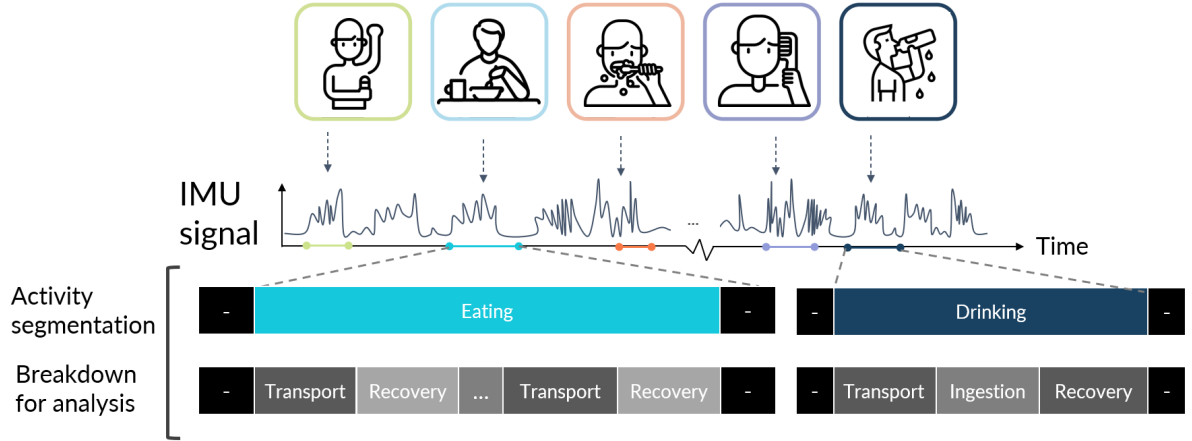


Figure 5.1: Two-stage activity segmentation

participant. For this, we used data from UA016, UA020, UA021, and UA025 in the train set, and data from GG98 and GG102 in the test set.

In the second setting, we trained an individualized model using only data from GG98 and GG102 using a leave-one-run-out approach. This setting is also known as the *individualized* setting. Each training fold was made up of two runs in condition A, two runs in condition B, and one run in condition C. Each test run consisted of one run in condition A and one run in condition B.

We made no changes to the hyperparameters from the multi-sensory dataset model except for adjusting the learning rate to achieve a more optimal learning behavior. We could predict our new tasks with only two examples of each non-negative class in the training set and one for validation. We did not include data augmentations. Only the forearm accelerations and gyroscope data were used as input. While the activity segmentation model was trained on complete runs, the meal segmentation model was only trained on data beginning five seconds before the eating activity and ending five seconds after the drinking activity.

Table 5.1 presents results for each of the settings for the activity segmentation model. The results were much better for the individualized models. For subject GG98, we reached F1 scores of 0.87 for the ADL segmentation and 0.71 for the meal segmentation task. For subject GG102, we obtained F1 scores of 0.88 for the ADL segmentation and 0.82 for the meal segmentation task. These performances come from very different results across the six runs used in cross-validation, with accuracies as low as 74% and as high as 96% for ADL segmentation depending on the run. Confusion matrices for the individualized models are shown in Figures 5.2 and 5.3. We observe that the inter-class confusion is higher for the meal segmentation algorithm because tasks are shorter and next to each other. For example, lifting the cup while drinking is confused with tilting the container to drink. For such tasks, it could be interesting to assess the variation in the ground-truth when multiple people label the same data and compare it to the segmentation done by the algorithm, as part of the error might come from variations in labels.

Model	ADL segmentation		Meal segmentation	
	Accuracy	F1-score	Accuracy	F1-score
GG98 generalized	0.42	0.51	0.22	0.22
GG98 individualized	0.87	0.85	0.71	0.71
GG102 generalized	0.61	0.65	0.67	0.54
GG102 individualized	0.88	0.87	0.84	0.82

Table 5.1: Performance comparison of different models tested on GG98 and GG102.



		Confusion Matrix (recall)												
True Labels	Null	0.98	0	0	0	0	0	0	0	0	0	0	0	0
	Left brushing	0.57	0.41	0.02	0	0	0	0	0	0	0	0	0	0
	Left comb	0.09	0.1	0.7	0.01	0.11	0	0	0	0	0	0	0	0
	Left deodorant	0.27	0	0	0.73	0	0	0	0	0	0	0	0	0
	Left drinking	0.18	0	0.04	0	0.72	0.06	0	0	0	0	0	0	0
	Left eating	0.33	0	0	0	0.01	0.66	0	0	0	0	0	0	0
	Left stirring	0.46	0	0	0	0	0	0.54	0	0	0	0	0	0
	Right brushing	0.02	0	0	0	0	0	0	0.89	0	0	0	0.06	0.02
	Right comb	0.1	0	0	0	0	0	0	0	0.84	0.05	0	0	0
	Right deodorant	0.44	0	0	0	0	0	0	0	0	0.56	0	0	0
	Right drinking	0.44	0	0	0	0	0	0	0	0	0	0.56	0	0
	Right eating	0.77	0	0	0	0	0	0	0.23	0	0	0	0	0
	Right stirring	0.72	0	0	0	0	0	0	0	0	0	0	0	0.28
			Null	Left brushing	Left comb	Left deodorant	Left drinking	Left eating	Left stirring	Right brushing	Right comb	Right deodorant	Right drinking	Right eating
		Predicted Labels												

Figure 5.2: Individualized model GG98 ADL segmentation confusion matrix.

Confusion Matrix (recall)

True Labels	Null	0.78	0	0	0	0.03	0.01	0	0.01	0.04	0.06	0.07
	Left drinking drink	0	0.97	0.02	0.01	0	0	0	0	0	0	0
	Left drinking lift	0.03	0.23	0.74	0	0	0	0	0	0	0	0
	Left drinking lower	0.4	0.08	0	0.52	0	0	0	0	0	0	0
	Left eating lift	0.1	0	0	0	0.9	0.01	0	0	0	0	0
	Left eating lower	0.19	0	0	0.01	0.15	0.66	0	0	0	0	0
	Right drinking drink	0	0	0	0	0	0	0.83	0.1	0.07	0	0
	Right drinking lift	0.73	0	0	0	0	0	0.17	0.11	0	0	0
	Right drinking lower	0.08	0	0	0	0	0	0.29	0	0.63	0	0
	Right eating lift	0.61	0	0	0	0	0	0	0.01	0.01	0.33	0.04
	Right eating lower	0.51	0	0	0	0	0	0.03	0	0.02	0.34	0.1
	Predicted Labels	Null	Left drinking drink	Left drinking lift	Left drinking lower	Left eating lift	Left eating lower	Right drinking drink	Right drinking lift	Right drinking lower	Right eating lift	Right eating lower

Figure 5.3: Individualized model GG98 meal segmentation confusion matrix.



Confusion Matrix (recall)

True Labels	Null	Left brushing	Left comb	Left deodorant	Left drinking	Left eating	Left stirring	Right brushing	Right comb	Right deodorant	Right drinking	Right eating	Right stirring
Null	0.97	0	0	0	0	0	0.01	0	0	0	0	0	0
Left brushing	0.01	0.96	0.03	0	0	0	0	0	0	0	0	0	0
Left comb	0.23	0.1	0.62	0.01	0.04	0	0	0	0	0	0	0	0
Left deodorant	0.4	0	0	0.6	0	0	0	0	0	0	0	0	0
Left drinking	0.13	0	0	0	0.85	0.02	0	0	0	0	0	0	0
Left eating	0.34	0	0	0	0.01	0.65	0	0	0	0	0	0	0
Left stirring	0.39	0	0	0	0	0	0.61	0	0	0	0	0	0
Right brushing	0.07	0	0	0	0	0	0	0.92	0.01	0	0	0	0
Right comb	0.01	0	0	0	0	0	0	0.01	0.98	0	0	0	0
Right deodorant	0.31	0	0	0	0	0	0	0	0	0.69	0	0	0
Right drinking	0.43	0	0	0	0	0	0	0	0.09	0	0.48	0	0
Right eating	0.37	0	0	0	0	0	0	0	0	0	0	0.52	0.1
Right stirring	0.96	0	0	0	0	0	0	0	0	0	0	0	0.04
	Null	Left brushing	Left comb	Left deodorant	Left drinking	Left eating	Left stirring	Right brushing	Right comb	Right deodorant	Right drinking	Right eating	Right stirring
	Predicted Labels												

Figure 5.4: Individualized model GG102 ADL segmentation confusion matrix.

Confusion Matrix (recall)

True Labels	Null	Left drinking drink	Left drinking lift	Left drinking lower	Left eating lift	Left eating lower	Right drinking drink	Right drinking lift	Right drinking lower	Right eating lift	Right eating lower
Null	0.96	0	0.01	0	0	0.01	0	0	0	0.02	0.01
Left drinking drink	0.07	0.78	0.05	0.01	0.08	0	0	0	0	0	0
Left drinking lift	0.37	0.05	0.5	0	0.01	0.03	0	0	0	0	0.04
Left drinking lower	0.35	0.09	0	0.56	0	0	0	0	0	0	0
Left eating lift	0.2	0	0.05	0	0.73	0.02	0	0	0	0	0
Left eating lower	0.18	0	0	0	0.04	0.78	0	0	0	0	0
Right drinking drink	0	0	0	0	0	0	0.92	0.04	0.03	0	0
Right drinking lift	0.09	0	0	0	0	0	0.27	0.64	0	0	0
Right drinking lower	0.66	0	0	0	0	0	0.03	0	0.32	0	0
Right eating lift	0.52	0	0	0	0	0	0	0	0	0.47	0.01
Right eating lower	0.45	0	0	0	0	0	0	0	0	0.08	0.47
	Null	Left drinking drink	Left drinking lift	Left drinking lower	Left eating lift	Left eating lower	Right drinking drink	Right drinking lift	Right drinking lower	Right eating lift	Right eating lower
	Predicted Labels										

Figure 5.5: Individualized model GG102 meal segmentation confusion matrix.



5.2 Discussion

More data is required both to train and assess the performance of individualized models on stroke participants but these preliminary results are promising. The challenge will come from obtaining enough data to capture the variability in ADLs and avoid overfitting. Regarding the generalized models, the data from the four healthy participants did not generalize at all to the stroke participants. One reason could be that the post-stroke individuals were performing all motions much slower than the healthy individuals, or using compensatory strategies. Once more healthy and stroke data is acquired, generalized models should be tested again on stroke participants. However, healthy data might not generalize well to the large variance of stroke participants' patterns used during ADLs, even with more subjects.

Chapter 6

Kinematic disparities between healthy and post-stroke ADLs with IMU sensors

In this chapter, we focus on assessing the variations in the performance of particular upper-extremity functional ADLs, between healthy individuals and stroke patients. Evaluating the quality of motion post-stroke is necessary for understanding the sensorimotor capabilities of patients. Unlike most existing literature, which predominantly assesses post-stroke motion by having the stroke survivor perform structured tasks, we emphasize the importance of evaluating movement quality during unstructured tasks. Furthermore, our analysis utilizes IMU data, while most existing work relies on optical motion capture data.

6.1 Task segmentation methodology

To isolate motions of interest within the StrokeRehab dataset [31] activities, we utilized joint angles obtained from the Euler angles computed from the IMU signals. These angles were employed to construct a simplified 3D geometric model of the human arms, as illustrated in Figure 6.1.

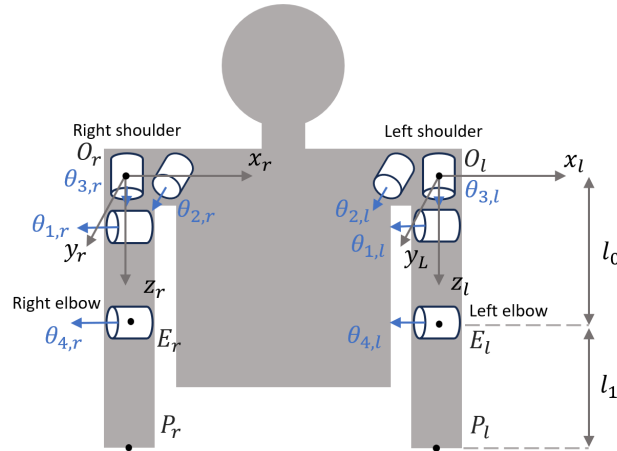


Figure 6.1: Simplified kinematic model of the arms used for visualization.



Here, we describe the direct geometric model of the left arm. Similar computations can be applied to the right arm.

$$\begin{aligned} \text{Left elbow position: } \overrightarrow{O_l E_l} &= K_{1,l} K_{2,l} K_{3,l} \overrightarrow{O_l E_{l0}} \\ \text{Left wrist position: } \overrightarrow{O_l P_l} &= K_{1,l} K_{2,l} K_{3,l} K_{4,l} \overrightarrow{O_l P_{l0}} \end{aligned}$$

where:

$$\begin{aligned} K_{1,l} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{1,l} & -s_{1,l} & 0 \\ 0 & s_{1,l} & c_{1,l} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & K_{2,l} &= \begin{bmatrix} c_{2,l} & 0 & s_{2,l} & 0 \\ 0 & 1 & 0 & 0 \\ -s_{2,l} & 0 & c_{2,l} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & K_{3,l} &= \begin{bmatrix} c_{3,l} & -s_{3,l} & 0 & 0 \\ s_{3,l} & c_{3,l} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ K_{4,l} &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & c_{4,l} & -s_{4,l} & l_0 s_{4,l} \\ 0 & s_{4,l} & c_{4,l} & (1 - c_{4,l})l_0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \overrightarrow{O_l E_{l0}} &= \begin{bmatrix} 0 \\ 0 \\ l_0 \\ 1 \end{bmatrix} & \overrightarrow{O_l P_{l0}} &= \begin{bmatrix} 0 \\ 0 \\ l_0 + l_1 \\ 1 \end{bmatrix} \end{aligned} \quad (6.1)$$

with $c_{i,l} = \cos \theta_{i,l}$ and $s_{i,l} = \sin \theta_{i,l}$ for $i = 1, 2, 3, 4$.

Through manual inspection of the IMU signals and derived angles, coupled with an understanding of the contextual motions, we confidently segmented activities of daily living for further analysis. Specifically, we focused on the motions associated with brushing teeth and drinking from a container.

6.2 Metrics for motion analysis of ADLs

The most common kinematic deficits for stroke patients during goal-directed motions are longer times to perform the motions, discontinuous spatial and temporal motion trajectories, and irregular movement patterns [72, 73, 74]. They also use compensatory movements (e.g., recruiting range of motion from other joints to compensate for a deficit in another joint) to help with completing tasks. Jayasinghe et al. (2021) have shown that stroke survivors tend to use more trunk compensation when reaching with their affected arm compared to their unaffected arm [75]. Wang et al. (2022) conducted a review of technologies used for compensation assessment and detection during upper extremity (UE) rehabilitation for stroke patients. The review highlights that wearable IMU sensors can be used for compensation assessment and detection during structured tasks [76].

Schwarz et al. (2019) give valuable recommendations for kinematic upper limb assessment post-stroke after reviewing existing work we used for our analysis [77]. Among these recommendations, they suggest that metrics should be selected based on clinimetric evidence and insights from motor control and that they should be hypothesis-driven.

We formulate a set of robust metrics to assess potential differences among healthy individuals, those with mild impairments from stroke, and those with moderate impairments. These metrics aim to provide a holistic view of the patient's quality of motion. We divide these metrics into three categories that cover the different aspects of motion that we identified: range of motion and mobility; temporal efficiency; and coordination and quality of movement.

Range of motion and mobility

Several studies highlight the importance of Range Of Motion (ROM) in stroke rehabilitation. Stroke results in a reduced ROM for joints on the paretic side [78]. Beebe and Lang (2019) showed that measurements of Active Range of Motion (AROM) of the upper extremity taken within one month after stroke could be used to predict upper extremity function after three months [79]. We investigate



mobility-related metrics mainly related to joint angle values. Stroke participants' motions are highly variable. Indeed, impairment might vary greatly depending on how the stroke occurred. Different stroke patients can reach different ROMs, and many tend to use other compensatory motions to overcome their impairments and still be able to perform the motions. Analyzing the amplitude of joints can reveal anomalies in use for certain activities.

Temporal efficiency

Temporal efficiency metrics relate to the effectiveness with which participants perform a movement. One underlying assumption is that because healthy participants do not struggle to do the activities, they might use less time and generate larger acceleration values. Possible metrics to investigate are total movement time, time to peak velocity, and velocity and acceleration values.

Coordination and quality of movement

After a stroke, patients tend to present a deficit in inter-joint coordination that can lead to segmented arm motions. During rehabilitation and recovery, there is evidence that patients regain this smoothness of motion [80]. To evaluate inter-joint coordination, we focus on smoothness computation and inter-joint cross-correlation. Cross-correlation measures the correlation between two-time series signals over time. The result is a correlation coefficient between -1 and 1 that indicates the strength and the direction of the relationship between the two signals. Smoothness of movement refers to the degree of continuity or lack of interruption in a motion. It assesses how closely a movement mirrors the pattern of starting with acceleration and then transitioning into deceleration. Melendez-Calderon et al. (2021) evaluated the validity of the different smoothness measures derived from IMU measurements [81]. Top approaches to measuring smoothness include Spectral Arc Length (SPARC) and log dimensionless jerk (LDLJ-V). It has been concluded that both measures should preferably be used with velocity measurements. They are prone to errors when applied to translational velocity reconstructed from acceleration due to drift. In addition, SPARC can be directly applied to rotational velocities, whereas LDLJ-V generates an error. SPARC [82] has been chiefly used in literature to quantify smoothness in point-to-point reaching motions in stroke. One of the significant benefits of using this measure is that it computes in the frequency domain, which means that the length of the motion doesn't impact the smoothness calculation. Other metrics, such as trajectory, velocity, and jerk, require adjustments due to changes in length, but this isn't the case for SPARC. The main idea behind SPARC computation is that less smooth movements have more frequency composition when approaching the signal with a Fourier series. Beck et al. (2018) compared the smoothness of walking using inertial sensors during walking in Parkinson's disease, computing SPARC from acceleration and gyroscope data [83]. Garcia et al. (2021) used SPARC for smoothness analysis of gait using IMUs [84]. Similarly to what they did, we first subtract the mean of the signal when computing smoothness and apply a cutoff frequency of 10 hertz. SPARC is calculated on the entire motion. Refai et al. (2021) obtained kinematic and clinical measurements in forty stroke participants during a reach-to-grasp movement in weeks 1 to 26 post-stroke [85]. The study revealed that the restoration of smoothness in a reaching task is correlated over time with the recovery from motor impairment as evaluated by the UEFMA. Until five weeks post-stroke, both SPARC and UEFMA scoring improved significantly.

The SPARC formula [81] is given by:

$$\lambda_S^v(v) \triangleq - \int_0^{\omega_c} \sqrt{\left(\frac{1}{\omega_c}\right)^2 + \left(\frac{d\hat{V}(\omega)}{d\omega}\right)^2} d\omega \quad (6.2)$$

where $\hat{V}(\omega) = \frac{V(\omega)}{V(0)}$ and $V(\omega) = |\mathfrak{F}(\|v(t)\|^2)|$.

$\mathbf{v}(t)$ represents the angular velocity of a movement in the time domain, $\mathfrak{F}(\cdot)$ is the Fourier transform operator, ω_c is an adaptative cutoff frequency.



With this definition, motions with low SPARC values are unsmooth compared to those with high SPARC values.

6.3 Brushing teeth post-stroke motion analysis

Using the method described in Section 6.1, we were able to extract with high confidence 86 samples (583 seconds) of healthy subject brushing coming from 18 different subjects, 84 samples (404 seconds) of mild (FMA 26-52) subject brushing from 19 subjects, and 98 samples (918 seconds) of moderate (FMA 26-52) subject brushing from 21 subjects.

Motivation for using the toothbrushing ADL for analysis of motor impairment in stroke.

In [86], Inada et al. evaluate the arm joint motion during toothbrushing motion. Their study highlights that the toothbrushing motion typically falls within the frequency range of 3.9 to 4.5 Hz, that it is a consistent and synchronous motion across all arm joints, and that the elbow plays a pivotal role in maintaining stability during brushing. Simkins et al. (2013) [87] investigated the effect of rhythmic motions on joint synergies between shoulder and elbow rotation after stroke at different speeds. It was concluded that people with hemiparesis move their arms at a lower frequency when asked to move as fast as possible. They also use smaller amplitudes, and the more the speed of the rhythmic motion increases, the more synergies are observed. This motivates the analysis of brushing teeth ADL for analyzing motor impairment in stroke.

In the following, we showcase, for a selection of selected metrics, the comparisons of their distribution for healthy, mildly impaired, and moderately impaired subjects. More evaluated metrics can be found in Table B.3.

Arm rotational speed when brushing teeth

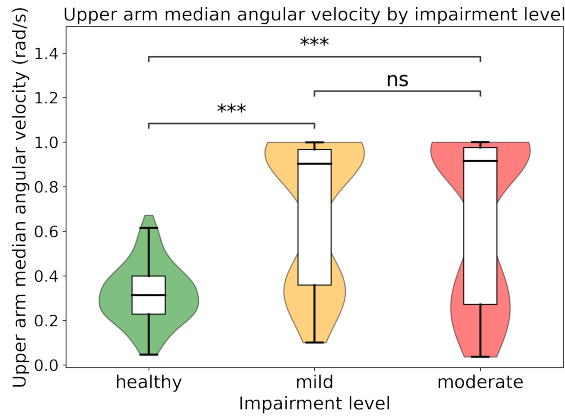


Figure 6.2: Brushing teeth median upper arm acceleration comparison between healthy, mild, and moderate subjects.

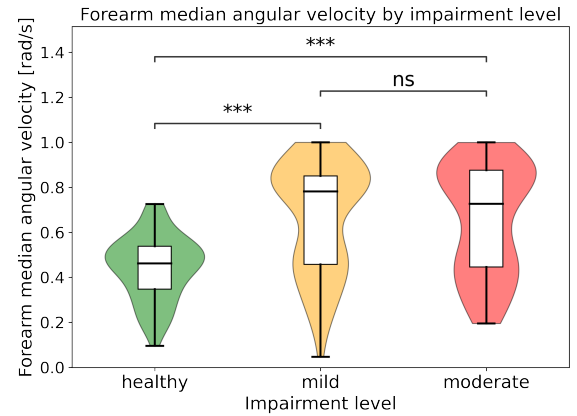


Figure 6.3: Brushing teeth median forearm acceleration comparison between healthy, mild and moderate subjects.

We conducted an Analysis of Variance (ANOVA) to assess the overall group differences in upper arm angular velocity (see Figure 6.2) during the brushing teeth activity ($F = 42.57, p < 0.0001$), revealing a large effect size ($\eta^2 = 0.243$). Post-hoc t-tests with Bonferroni correction confirmed significant differences between healthy and stroke participants:

- Healthy vs. Mild: significant ($p < 0.0001, t = -9.95$).
- Mild vs. Moderate: not significant ($p = 1.00, t = 0.75$).



- Healthy vs. Moderate: significant ($p < 0.0001$, $t = -8.08$).

Healthy participants (mean=0.32rad/s, SD=0.13rad/s) tend to use smaller upper arm angular velocities than mild (mean=0.69rad/s, SD=0.32rad/s) and moderate (mean=0.65rad/s, SD=0.36rad/s).

Similarly, for the forearm (Figure 6.3), ANOVA showed significant results ($F = 30.41$, $p < 0.0001$) with a large effect size ($\eta^2 = 0.187$). Post-hoc t-tests with Bonferroni correction revealed:

- Healthy vs. Mild: significant ($p < 0.0001$, $t = -7.45$).
- Mild vs. Moderate: not significant ($p = 1$, $t = 0.35$).
- Healthy vs. Moderate: significant ($p < 0.0001$, $t = 7.16$).

As for the upper arm, healthy participants (mean=0.44rad/s, SD=0.14rad/s) used smaller forearm median angular velocities than mild (mean=0.67rad/s, SD=0.24rad/s) and moderate (mean=0.66rad/s, SD=0.24rad/s).

Individuals who have had a stroke tend to rotate their arms at a higher velocity when brushing their teeth, in comparison to healthy individuals. We hypothesize that this increase in angular velocity may be a compensatory mechanism, which reflects their adaptation to overcome the reduced motor control caused by the stroke. For instance, if the wrist is unable to move the toothbrush, they rotate their arm more to compensate for the lack of movement.

Arm acceleration when brushing teeth

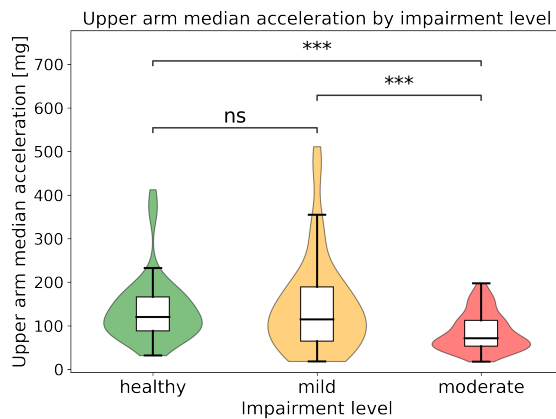


Figure 6.4: Brushing teeth median upper arm acceleration comparison between healthy, mild, and moderate subjects.

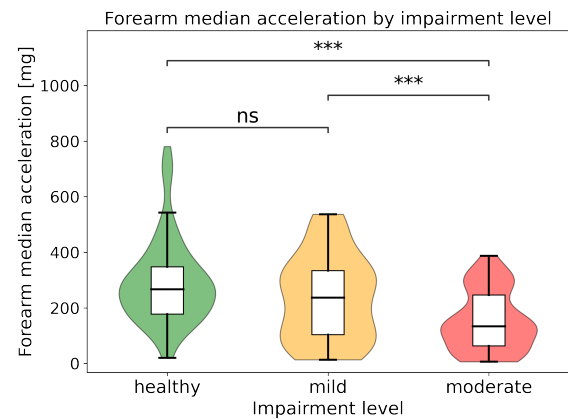


Figure 6.5: Brushing teeth median forearm acceleration comparison between healthy, mild and moderate subjects.

ANOVA used to assess the overall group differences in upper arm acceleration (see Figure 6.4) during the brushing teeth activity ($F = 15$, $p < 0.0001$), revealed a medium effect size ($\eta^2 = 0.102$). Post-hoc t-tests with Bonferroni correction confirmed significant differences between groups:

- Healthy vs. Mild: not significant ($p = 1$, $t = -0.52$).
- Mild vs. Moderate: significant ($p < 0.0001$, $t = 4.86$).
- Healthy vs. Moderate: significant ($p < 0.0001$, $t = 5.71$).

Moderately impaired patients present a smaller median upper arm acceleration (mean=85.8mg, SD=42.8mg) than mild (mean=143.9mg, SD=108.3mg) and healthy (mean=136.43mg, SD=74.5mg).

Similarly, for the forearm (Figure 6.5), ANOVA showed significant results ($F = 23.86$, $p < 0.0001$) with a large effect size ($\eta^2 = 0.187$). Post-hoc t-tests with Bonferroni correction revealed:

- Healthy vs. Mild: not significant ($p = 0.06$, $t = 2.37$).



- Mild vs. Moderate: significant ($p = 0.0001, t = 4.28$).
- Healthy vs. Moderate: significant ($p < 0.0001, t = 7.26$).

Moderately impaired patients use lower median forearm acceleration (mean=152.7mg, SD=148.6mg) than mild (mean=235.3mg, SD=151.8mg) and healthy (mean=290.4mg, SD=148.6mg).

Overall, stroke participants use less acceleration when brushing their teeth. This can be due to muscle weakness, coordination loss, joint stiffness, sensory impairments, or cognitive issues.

Joint angle amplitudes when brushing teeth

We also investigated the differences in joint angle amplitudes during brushing teeth. We hypothesized that stroke survivors might find it challenging to elevate their arms, in contrast with arm elevation used by healthy subjects. We thus analyzed the range of shoulder abduction (Figure 6.6) and shoulder flexion (Figure 6.7) angles being used by stroke and healthy participants.

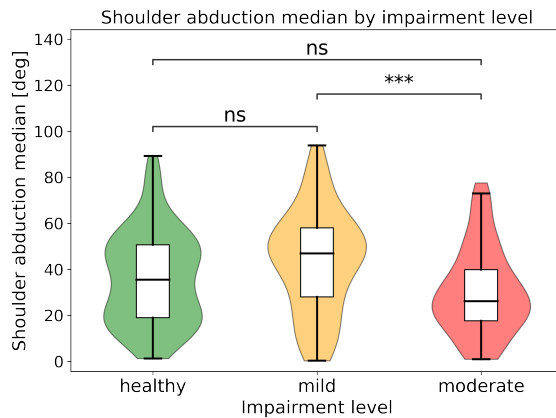


Figure 6.6: Brushing teeth median shoulder abduction comparison between healthy, mild, and moderate subjects.

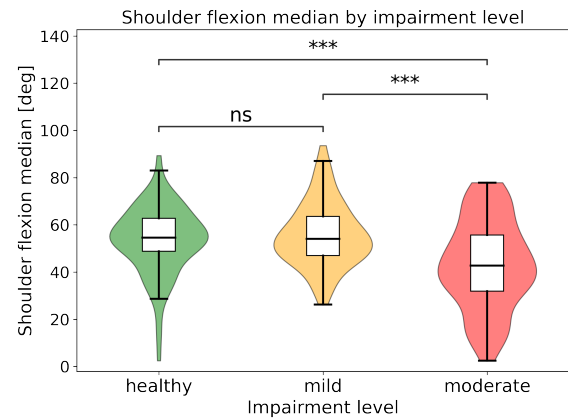


Figure 6.7: Brushing teeth median shoulder flexion comparison between healthy, mild and moderate subjects.

For shoulder abduction, ANOVA highlighted significant differences between groups ($F = 9.69, p = 0.0001$), with a medium effect size ($\eta^2 = 0.0682$). Post-hoc t-tests with Bonferroni correction only confirmed significantly lower abduction joint angle used by moderate compared to mild.

- Healthy vs. Mild: not significant ($p = 0.076, t = -2.26$).
- Mild vs. Moderate: significant ($p = 0.0001, t = 4.38$).
- Healthy vs. Moderate: significant ($p = 0.11, t = 2.08$).

On average, moderately impaired participants use a median shoulder abduction of 30.4 degrees (SD=18.4 deg), mild 43.6 degrees (SD=22.16 deg), and healthy participants 36.3 degrees (SD=19.9 deg).

Regarding flexion, ANOVA also confirmed differences ($F = 16.96, p < 0.0001, \eta^2 = 0.113$), and post-hoc showed that moderately impaired subjects use significantly lower shoulder flexion angles (mean=43.1 deg, SD=18.7 deg) than mild (mean=55.8 deg, SD=13.5 deg) and healthy (mean=53.6 deg, SD=14.4 deg).

- Healthy vs. Mild: not significant ($p = 0.87, t = -1.06$).
- Mild vs. Moderate: significant ($p < 0.0001, t = 5.18$).
- Healthy vs. Moderate: significant ($p = 0.0001, t = 4.20$).

Paretic/Non-paretic arm brushing asymmetry



Among all the participants, only one brushed his teeth changing the arm brushing. A plot showing the evolution of the joint angles for a mild subject brushing first with its paretic arm and then switching the arm brushing to brush with the non-paretic arm is presented in Figure 6.8. We observed that while brushing with the non-impaired arm, shoulder rotation and shoulder flexion angles follow a sinusoidal pattern; however, when brushing with the impaired arm, the angles are much less regular.

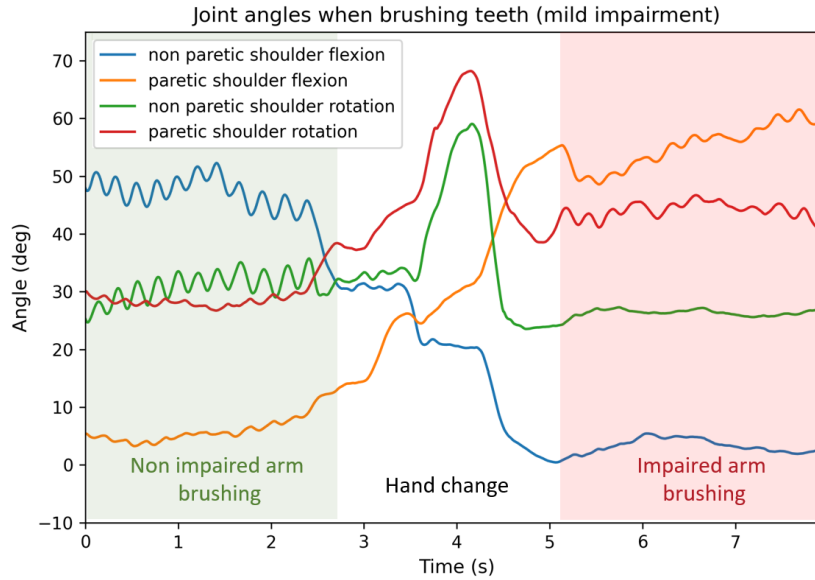


Figure 6.8: Paretic/non-paretic brushing motion asymmetry in shoulder joint angles

Conclusions on stroke vs. healthy participants brushing motion

Our analyses showed that stroke participants use less accelerations, higher rotational velocities, and smaller shoulder elevation when brushing their teeth. We found no significant difference in elbow joint angles being used and in the frequency of the brushing motion. The amplitude of the motion is also larger for healthy individuals compared to post-stroke, as shown by the higher median absolute deviation of both shoulder flexion, abduction and rotation.

Brushing teeth can contribute to evaluating impairment levels during activities of daily living, as it is a challenging and constrained activity that requires mobility and coordination in the upper extremities. While complex ADLs might be more efficient for highlighting differences between healthy and stroke participants, severely impaired participants would not be able to do them, which is a downside.



6.4 Drinking post-stroke motion analysis

6.4.1 Decomposition of the drinking ADL

The drinking motion during which the hand is in contact with an open container can be decomposed into three distinct phases based on their kinematic properties, as described by Murphy's phase distribution:

1. Transport phase: This phase involves elevating the liquid container from the resting position to the mouth. The primary motions are elbow flexion and shoulder abduction/flexion. As this phase requires dynamically moving an object, the kinematics can be characterized as dynamic.
2. Ingestion phase: The goal is to stabilize the container while sipping liquid. Minor adjustments such as forearm rotation or shoulder abduction/flexion may be used to tilt the container precisely. Compared to the transport phase, this phase emphasizes stability over dynamic movement.
3. Recovery phase: In the recovery phase, the container is lowered back to its original position. This is accomplished primarily through elbow extension and shoulder flexion/abduction reduction.

Figure 6.9 illustrates this decomposition, indicating key events that mark the start and end of each phase.

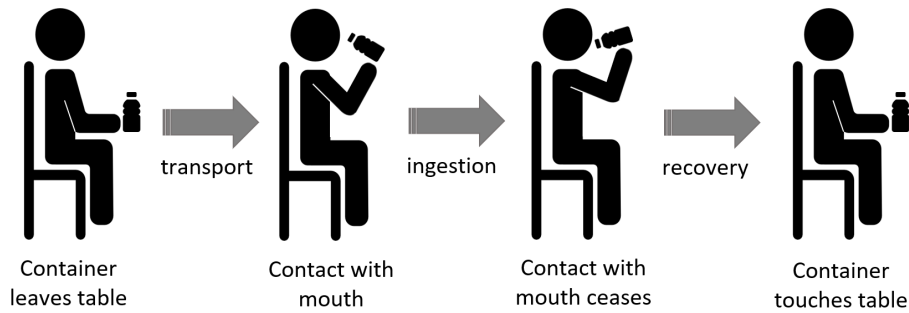


Figure 6.9: Decomposition of the drinking task.

By analyzing each drinking phase separately based on their distinct kinematic properties, we can develop a more scientifically rigorous understanding of the biomechanics underlying this everyday task. The drinking ADL has been well studied, mainly in structured experiments isolating each phase and with motion capture systems. It has been much less studied using wearable sensors in unstructured settings.

In [88], Murphy et al. evaluated the change of a set of kinematical parameters describing movement time, velocity, strategy, interjoint coordination, and compensatory movement between stroke and healthy participants during drinking task using motion capture. They found that the number of movement units, total movement time, and peak angular velocity of the elbow were the best features to discriminate between healthy, mild, and moderate groups, with the number of movement units being the best indicator of motion quality. They also showed that time is correlated with number of movement units.

Kim et al. (2014) [89] conducted a study identifying kinematic differences between 16 hemiplegic and 32 healthy participants during drinking phases. The hemiplegic group took longer to achieve the drinking task, taking extra care when lifting the cup to avoid spilling water and caring less during recovery. Interestingly, hemiplegic participants used more extensive abduction in all phases of drinking and smaller flexion in the shoulder joint for the transporting and ingestion phases.



6.4.2 Analysis

We analyze both dynamic phases of the drinking task with the same set of metrics. Using the segmentation method described in Section 6.1, we could confidently isolate the transport, ingestion, and recovery phases of drinking. For each stage, we could isolate 70 samples from 15 mildly impaired participants, 78 samples from 18 moderately impaired participants, and 86 samples from 18 healthy subjects, totaling 234 drinking samples analyzed. As for the toothbrushing ADL, we only discuss here a subset of the analyzed metrics, more detailed analysis can be found in Tables B.1 and B.2.

Smoothness

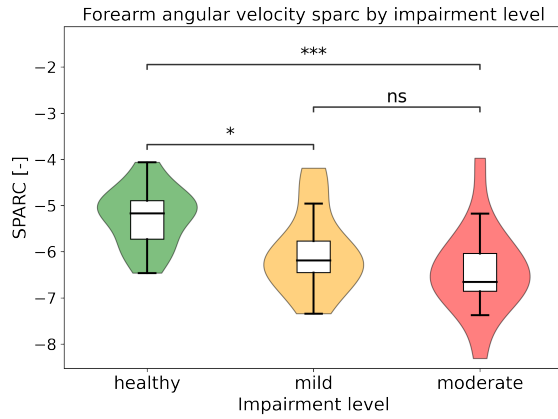


Figure 6.10: Drinking **transport** forearm angular velocity based SPARC distributions of healthy, mild, and moderate subjects.

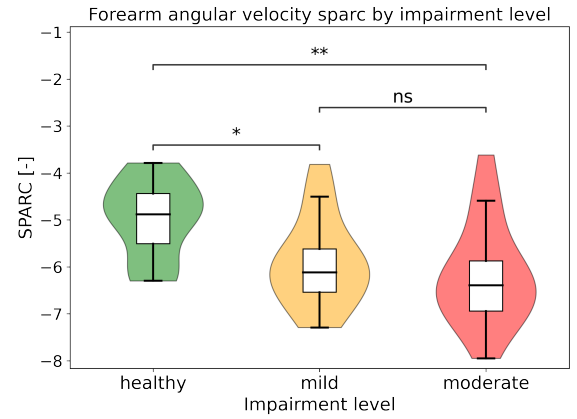


Figure 6.11: Drinking **recovery** forearm angular velocity based SPARC distributions of healthy, mild, and moderate subjects.

Wrist angular speed SPARC during the drinking transport phase distributions are represented in Figure 6.10. We conducted an Analysis of Variance (ANOVA) to assess the overall group differences ($F = 8.93, p = 0.0005$), revealing a large effect size ($\eta^2 = 0.27$). Post-hoc t-tests with Bonferroni correction confirmed significant differences:

- Healthy vs. Mild: significant ($p = 0.020, t = 2.91$).
- Mild vs. Moderate: non significant ($p = 0.71, t = 1.21$).
- Healthy vs. Moderate: significant ($p = 0.0005, t = 4.26$).

Healthy participants (mean=-5.25, SD=0.62) present higher transport wrist angular velocity SPARC than mild (mean=-6.01, SD=0.84) and moderate (mean=-6.40, SD=0.92).

Similarly, for the recovery phase (Figure 6.11), ANOVA showed significant results ($F = 7.62, p = 0.001$) with a large effect size ($\eta^2 = 0.24$). Post hoc t-tests with Bonferroni correction revealed:

- Healthy vs. Mild: significant ($p = 0.014, t = 3.04$).
- Mild vs. Moderate: non significant ($p = 1, t = 0.73$).
- Healthy vs. Moderate: significant ($p = 0.002, t = 3.71$).

Healthy participants (mean=-4.96, SD=0.80) present higher recovery wrist angular velocity SPARC than mild (mean=-5.90, SD=0.92) and moderate (mean=-6.16, SD=1.06).

Figures 6.10 and 6.11 show that the healthy group exhibits significantly smoother transport and recovery motion when drinking compared to the mildly and moderately impaired stroke participants, which is in line with the literature.



Time efficiency during transport and recovery phase

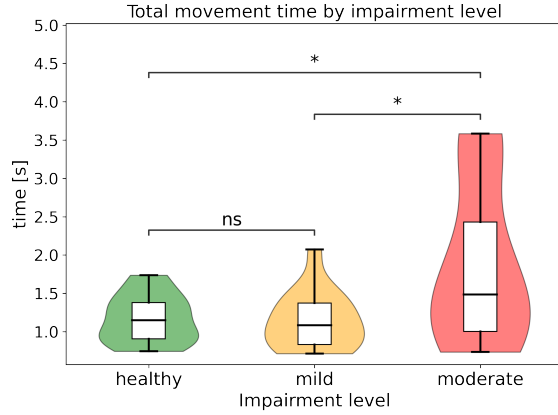


Figure 6.12: Drinking **transport** movement time distributions of healthy, mild, and moderate subjects.

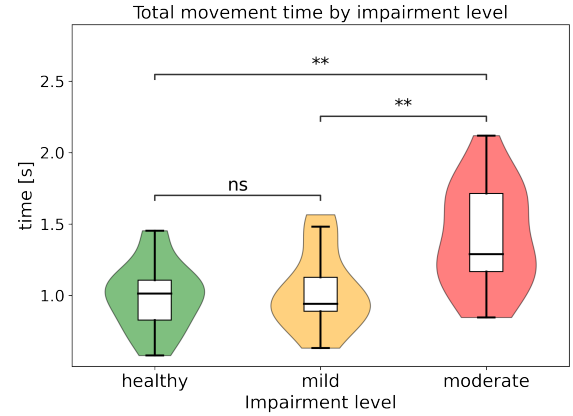


Figure 6.13: Drinking **recovery** mean movement time of healthy, mild, and moderate subjects.

We evaluated group differences for total transport movement time (Figure 6.12), ANOVA showed significant results ($F = 6.35, p = 0.004$) with a large effect size ($\eta^2 = 0.213$). Post-hoc t-tests with Bonferroni correction revealed:

- Healthy vs. Mild: not significant ($p = 1, t = 2.68$).
- Mild vs. Moderate: significant ($p = 0.047, t = 2.56$).
- Healthy vs. Moderate: significant ($p = 0.032, t = 2.71$).

Moderately impaired subjects (mean=1.87s, SD=1.02s) perform the transport phase using more time than mildly impaired (mean=1.14s, SD=0.36s) and healthy participants (mean=1.17s, SD=0.30s).

Similarly, for the recovery phase (Figure 6.13), ANOVA resulted in $F = 10.03, p = 0.0002, \eta^2 = 0.299$ and post-hoc:

- Healthy vs. Mild: not significant ($p = 1, t = 0.4$).
- Mild vs. Moderate: significant ($p = 0.008, t = 3.26$).
- Healthy vs. Moderate: significant ($p = 0.001, t = 3.98$).

As for the transport phase, the moderate group (mean=1.41s, SD=0.35) performed the recovery phase slower than the healthy (mean=0.99s, SD=0.23s) and mild (mean = 1.03, SD=0.26).

We conclude that moderately impaired patients tend to perform the motions much slower than mildly impaired and healthy participants, with more significant intra-group differences.

6.5 Conclusion on post-stroke upper-extremity motion analysis within ADLs

In conclusion, our study on the drinking and tooth-brushing motions gives us important insights into how people who have had a stroke compare to those who have not. Although specific measurements help to separate the two groups, there is still some overlap between the distributions of the different groups.

The metrics investigated (see Tables B.3, B.1 and B.2) include some descriptive statistics such as medians, 95th percentile, and median absolute deviation calculated on the forearm and upper arm



norms of acceleration and angular velocity. These descriptive statistics are robust versions of mean, peak, and standard deviation. For the toothbrushing ADL, we also looked at shoulder abduction, rotation, and flexion as most of the motion comes from the shoulder. Then, for drinking tasks, we added total movement time and SPARC to measure smoothness and cross-correlation between forearm and upper arm angular velocities and accelerations. These additional task-specific metrics have been shown to effectively distinguish between the movements of participants who have had a stroke and healthy participants. Comparison between healthy, mildly impaired, and moderately impaired subjects can be found in Appendix B.

Our findings suggest stroke survivors are likelier to present heavy tails in bio-mechanical metrics that translate a motion's quality aspect. Looking at specific aspects like SPARC and movement time, we notice interesting differences between stroke and healthy participants. While some stroke survivors move similarly to healthy individuals, many show less smooth motions and take more time to finish tasks. This highlights the fact that the experiences of stroke survivors can vary widely.

It is crucial to recognize the limitations of our study, like the limited range of motions and instances we analyzed. Future research with a broader set of motions and a larger dataset will help us better understand the subtleties of how motor skills play out in stroke survivors during activities of daily living.

Chapter 7

The motion healthiness score: A data-driven framework for naturalistic motion quality assessment

While UE motor impairment assessments like the UEFMA or the ARAT provide valuable information about upper extremity motor function, that can help treatment planning and track rehabilitation progress, they do not capture patients' capacity to perform specific functional tasks. For example, Murphy et al.[90] showed that kinematic movement performance measures obtained during a drinking task, such as movement time, smoothness, and angular velocities, correlate much more with activity capacity than with actual impairment level obtained through UEFMA and ARAT. Assessing impairment in a functional setting holds considerable promise to understand activity capacity better.

In an attempt to replicate the FMA score from wearable sensors, Yu et al. (2016) employed two accelerometers and seven flex sensors to monitor the function of the upper limb [91]. The proposed quantitative FMA model based on wearable sensor data can precisely predict the FMA scores, with a coefficient of determination of up to 0.917. Chaeibakhsh et al. (2016) used accelerometer and gyroscope data to score UEFMA using machine learning algorithms [92]. Although these works offer a good opportunity to do assessments at home, without the need for a therapist, impairment is still evaluated on very structured tasks. Oubre et al. (2020) [93] proposed an approach to estimate upper-limb impairment in stroke survivors by using two wearable inertial sensors on the wrist and sternum and evaluating on two minutes of continuous random voluntary upper limb movements spanning the entire range of motion. They achieved a root mean square error of 18.2% for 23 stroke survivors. Vanmechelen et al. (2023) conducted a scoping review on using wearable sensors to assess movement disorders during upper limb tasks. The review found that most of the assessments were carried out during structured tasks and identified commonly computed metrics that are used to identify different kinds of impairments [94].

In Chapter 6, we observed distinct distributions between the Healthy, Mild, and Moderate participant groups for the identified set of robust wearable-IMU-based metrics. This suggests that valuable information about motion quality is contained within the IMU data during specific constrained ADL segments. Within this chapter, our primary objective is to develop a method to incorporate identified metrics into a framework for the assessment of naturalistic motion quality. We aim to calculate a Motion Healthiness Score (MHS) for each movement by integrating various metrics related to mobility, time efficiency, and overall motion characteristics.



The fundamental concept behind the MHS we develop is to gauge the disparity between the motion under evaluation and the same motion executed by healthy subjects. This approach is based on the assumption that the standard for high-quality of motion is established by the movements of healthy individuals. We take into account the substantial variability in stroke impairments among individuals: some stroke participants may execute certain motions with proficiency while struggling with others. Furthermore, a fundamental principle of stroke rehabilitation is to gradually align movements with those that healthy individuals would typically perform, and this score measures the quality of the alignment. The proposed score has convenient applications. It could be communicated to therapists and patients to monitor rehabilitation and gauge progress in motion performance. Therapists could use this information to emphasize training for motions with lower MHS to optimize recovery.

7.1 Methods and results

In this Section, we explain the derivation of the MHS and compute the MHS for StrokeRehab participants in three motions: brushing teeth, the transport phase of drinking (lifting the container), and the recovery phase of drinking (lowering the container).

Figure 7.1 shows the principal steps of the MHS framework. The first step is to pre-segment a motion of interest, this can be done either manually or using our activity segmentation algorithm (see Chapter 3). Then, we compute a set of robust metrics, as defined in Chapter 6. Subsequent steps involve reducing the dimensionality of the metrics using PCA and computing the distance between each motion and healthy references for that motion in the principal component space. More detailed explanations of these steps will follow in the following sections.

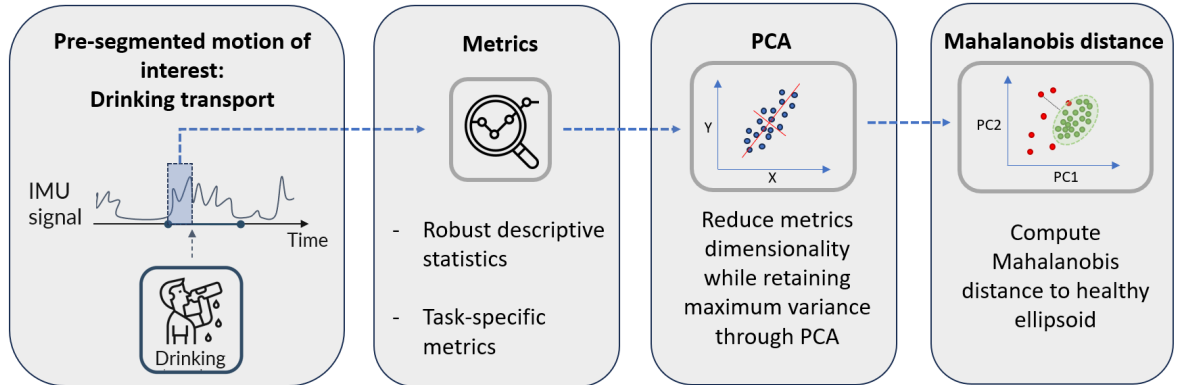


Figure 7.1: Motion healthiness score (MHS) framework

7.1.1 PCA dimensionality reduction

In Chapter 6, we identified a set of robust descriptive statistics and task-specific metrics, that are interpretable, and allow for Healthy, Mild, and Moderate group differentiation. We first compute these metrics for each pre-segmented motion.

Subsequently, employing Principal Component Analysis (PCA), we reduce the dimensionality of the computed metrics, optimizing the retained variance. A desirable property of PCA is its orthogonality: principal components are orthogonal to each other, which makes them linearly independent. Each principal component captures a non-redundant aspect of the data.

To apply PCA, we first compute the covariance matrix Σ :



$$\Sigma = \frac{1}{n-1}(X - \bar{X})^T(X - \bar{X}) \quad (7.1)$$

with X the data matrix, $\bar{X}$ the mean of each feature and n the number of features.

We then compute the eigenvalues λ_i of the covariance matrix by solving the following:

$$\det(\Sigma - \lambda I) = 0 \quad (7.2)$$

and solve for the eigenvectors $\mathbf{v}_i$ by substituting each eigenvalue into:

$$(\Sigma - \lambda_i I)\mathbf{v}_i = 0 \quad (7.3)$$

We then sort the eigenvalues in descending order and select all $\lambda_i > 1$ following the Guttman-Kaiser criterion [95]. This criterion is based on the argument that no component explaining less than the variance of an original variate represents a significant source dimension. Although it may not be optimal in all situations, it is a widely-spread, easy-to-use method for selecting the number of principal components. We select the corresponding eigenvectors to form the projection matrix W .

$$W = [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k]$$

Where k is the number of eigenvalues greater than 1.

Finally, we can project the data matrix X in the subspace of principal components defined by the eigenvectors:

$$Y = XW \quad (7.4)$$

Figure 7.2 shows the scree plots for the three tasks we analyze. The Guttman-Kaiser criterion keeps the six first principal components for the brushing teeth activity, five for the drinking transport activity, and five for the drinking recovery activity, leading to respectively 88%, 84%, and 86% of explained variance retained.

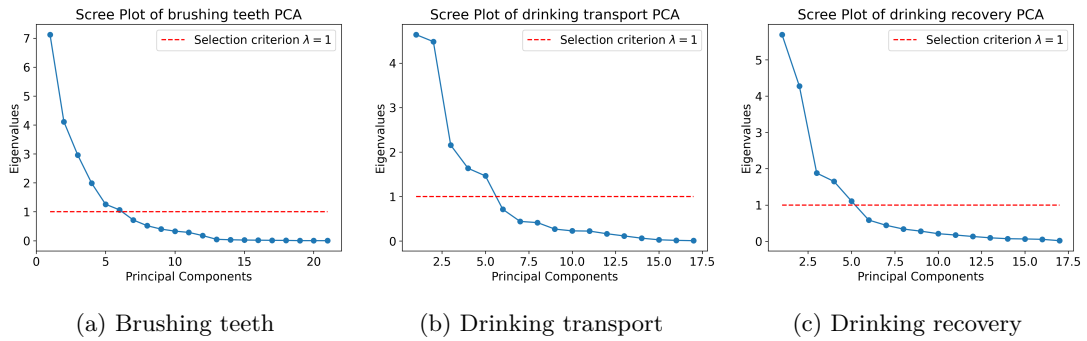


Figure 7.2: Scree plots for brushing teeth, drinking transport and recovery PCA



7.1.2 Mahalanobis distance to healthy data

The Mahalanobis distance [96] measures the distance between a point and a distribution under the assumption that the data conforms to a multivariate normal distribution. This distance metric is more relevant than the Euclidean distance because it incorporates the covariance structure of the data, allowing for considering correlations between variables. It is computed as:

$$D_M^2 = (x - m)^T \cdot C^{-1} \cdot (x - m) \quad (7.5)$$

Where D_M^2 is the square of the Mahalanobis distance, x is the observation vector, m is the mean of each variable, and C is the covariance matrix of the variables.

We use the Mahalanobis-MCD distance, which is known as a robust criterion for outlier detection. Instead of computing the mean and covariance matrix of the whole data, it uses the minimum covariance determinant (MCD) method to estimate the multivariate mean and covariance matrix using a subset of the data so that we minimize the influence of anomalies on their computations. Following the recommendations in [97], we use a support fraction of 0.75 when calculating the MCD. We use the Mahalanobis-MCD distance to compute the distance from sample motions to the distribution of healthy in the principal components subspace. For the sake of simplicity, we will refer to the Mahalanobis-MCD distance as Mahalanobis distance going forward.

The square of the Mahalanobis distance can be approximated by a Chi-square distribution, with a degree of freedom equal to the number of principal components. We use the quantile value 0.95 as a threshold to define the limit of what is considered healthy data. Any point with a Mahalanobis distance superior to the threshold value set using the Chi-Square distribution is regarded as an outlier. Any point within the threshold is given a maximum MHS, meaning the motion is considered healthy. To give an intuition on the method, we show in Figures 7.3, 7.4 and 7.5 the projection of metrics in the 3D space defined by the three first principal components, with, in green the ellipsoid of healthy data defined with this threshold. We observe that most of the healthy data fall into this ellipsoid, while mild and moderate samples are more dispersed. A black dot marks points outside of the ellipsoid. We refer to these points as outliers.

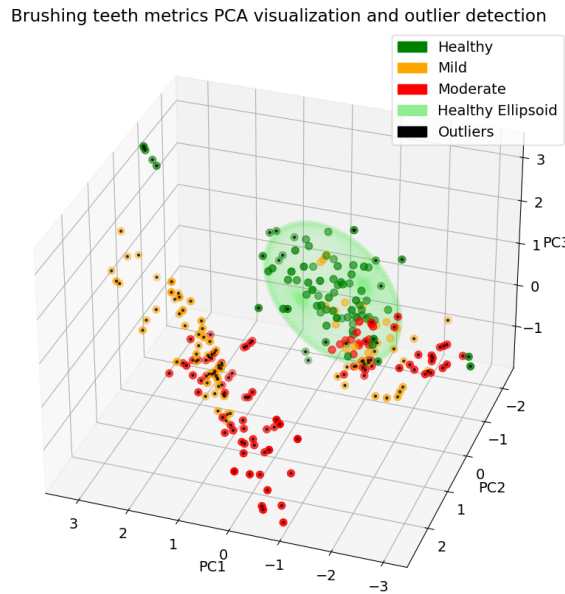


Figure 7.3: Brushing teeth PCA projection (3D)



Drinking transport PCA visualization and outlier detection

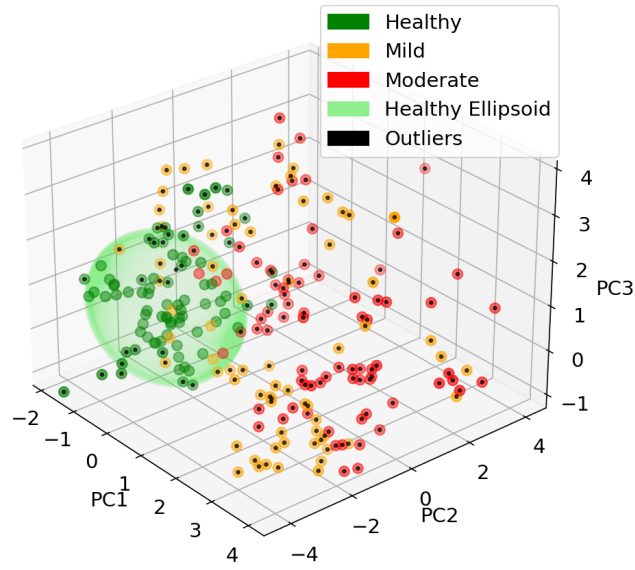


Figure 7.4: Drinking transport PCA projection (3D)

Drinking recovery PCA visualization and outlier detection

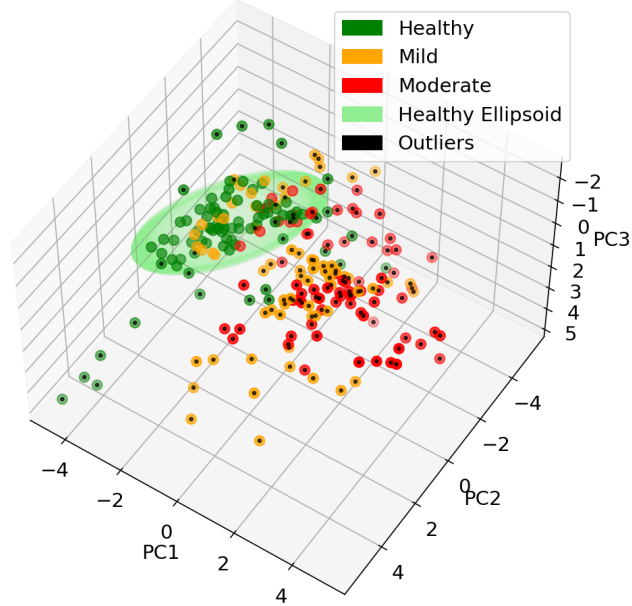


Figure 7.5: Drinking recovery PCA projection (3D)

Figures 7.6, 7.8 and 7.8 respectively show the Mahalanobis distance to the center of the healthy ellipsoid for both healthy and stroke participants. The green and red dotted lines, indicating distance thresholds, serve for the future calculation of the motion healthiness score from the Mahalanobis distance to the healthy data.

When brushing teeth, healthy individuals' motions have a mean Mahalanobis distance to the center of



the healthy ellipsoid of 3.22 (SD=1.69), while for post-stroke participants, the mean distance is 6.33 (SD=2.47).

Similarly, for the transport phase of drinking, the mean Mahalanobis distance for healthy motions is 2.99 (SD=1.93) and for stroke, 7.46 (SD=6.06). For the recovery phase, the mean distance is 2.99 (SD=1.97) for healthy subjects and 7.83 (SD=3.76) for stroke participants.

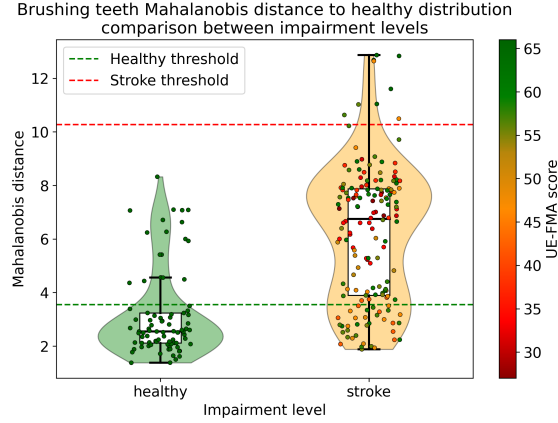


Figure 7.6: Brushing teeth Mahalanobis distance to healthy ellipsoid distribution.

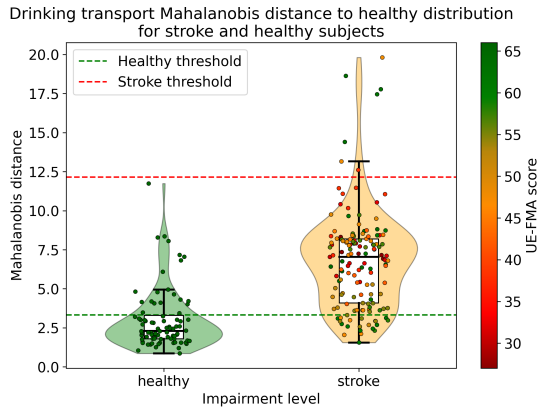


Figure 7.7: Drinking **transport** Mahalanobis distance to healthy ellipsoid distribution.

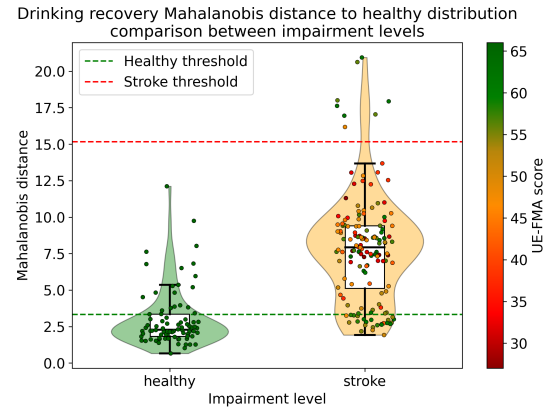


Figure 7.8: Drinking **recovery** Mahalanobis distance to healthy ellipsoid distribution.

Although we do not have a ground truth for how far a motion is to healthy data, we verify that our distance behaves well by demonstrating that mild group samples have a smaller distance to healthy data than moderate, by computing the correlation between the UE-FMA score and the distance for each sample, as well as by evaluating the variation of the resulting score from different samples of the same subject. We also evaluate the correlation between the scores obtained for each of the three tasks we examine.

We provide the Bhattacharyya distance [98] to measure overlap and convergence between distributions. The greater the Bhattacharyya distance is, the more dissimilar the two distributions are. The Bhattacharyya distance $D_{B_{1-2}}$ between two distributions 1 and 2 is computed by formula 7.6:



$$D_{B_{1-2}} = \frac{1}{8} (\mathbf{m}_1 - \mathbf{m}_2)^T \left(\frac{\mathbf{C}_1 + \mathbf{C}_2}{2} \right)^{-1} (\mathbf{m}_1 - \mathbf{m}_2) + \frac{1}{2} \log \left(\frac{\det \left(\frac{\mathbf{C}_1 + \mathbf{C}_2}{2} \right)}{\sqrt{\det(\mathbf{C}_1) \cdot \det(\mathbf{C}_2)}} \right) \quad (7.6)$$

where:

m_1 and m_2 are the means of the two distributions.

C_1 and C_2 are the covariance matrices of the two distributions.

Table 7.1 shows the Bhattacharyya distance between the distributions of principal components between the different activities. We notice that the healthy-to-moderate distance is more significant for all three activities than the healthy-to-mild distance, especially for the transport and recovery tasks, where it more than doubles.

	Brushing teeth	Drinking transport	Drinking recovery
Healthy to Mild D_B	1.60	0.92	1.16
Healthy to Moderate D_B	1.93	2.87	2.38
Mild to Moderate D_B	0.51	1.07	0.83

Table 7.1: Bhattacharyya distance between distributions of healthy, mild, and moderate groups in the principal component space for brushing teeth, drinking transport, and drinking recovery analysis.

We examined the Pearson correlation between brushing teeth, drinking transport, and drinking recovery distances to healthy data. The correlation coefficient between brushing and drinking transport Mahalanobis distances to the healthy ellipsoid is 0.59, indicating a moderate positive correlation. Brushing and recovery distances are correlated at 0.62, while transport and recovery distances show a notably strong correlation at 0.91. All distances are moderately or strongly correlated, with a very high correlation for each individual between the transport and recovery phases of drinking.

From the Mahalanobis distances computation, we define the MHS as follows. The best attainable MHS for a task is 10, obtained when the Mahalanobis distance to the healthy distribution is less than the distance defined by the 0.95 quantile of the chi-square distribution with a degree of freedom equal to the number of selected principal components. The worst score is 0 when the Mahalanobis distance to the healthy cluster is superior to the 0.95 quantile distance estimated with a Kernel Density Estimation (KDE) of stroke participants' distances. These two distance thresholds are shown in Figures 7.6, 7.7 and 7.8. Between these two values, the score is linearly interpolated using the piecewise linear function described in Equation 7.7:

$$\text{MHS} = 10 - (D_B - t_h) \times \frac{10}{t_s - t_h} \quad (7.7)$$

With t_h the 0.95 quantile of the chi-square distribution with a degree of freedom equal to the number of selected principal components, and t_s the 0.95 quantile distance estimated with a Kernel Density Estimation (KDE) of stroke participants' distances.

We use Silverman's bandwidth estimation for the KDE [99].

The mean Mahalanobis distance to the healthy distribution and the resulting MHS for each subject and each activity are reported in Table 7.2 and shown in Figures 7.9, 7.10 and 7.11.

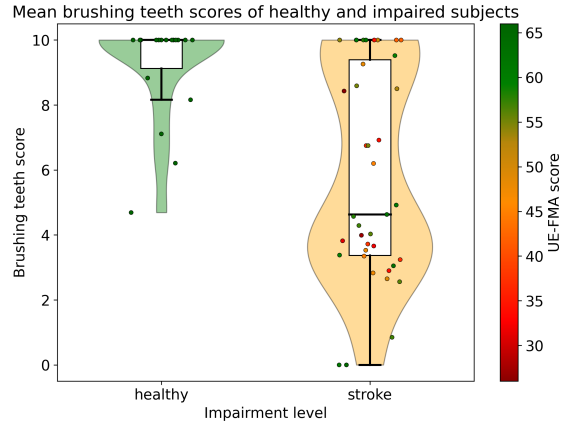
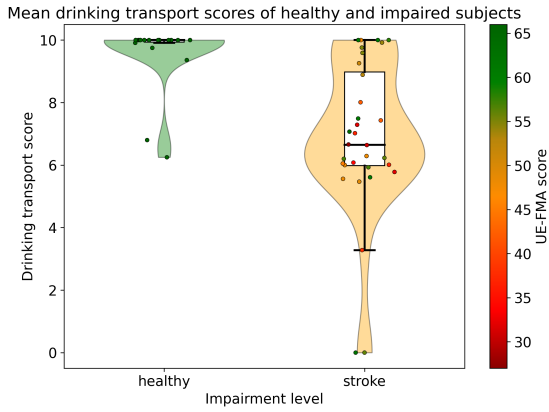
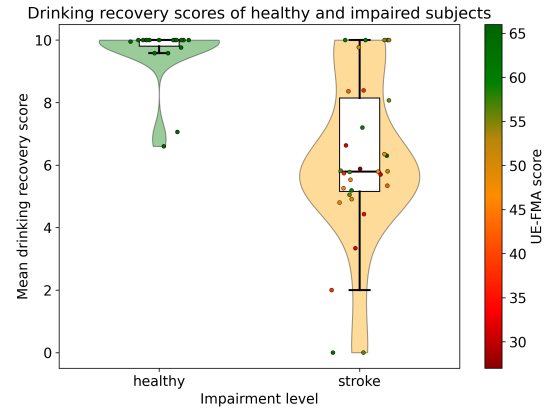


Figure 7.9: Brushing teeth mean MHS per subject.

Figure 7.10: Drinking **transport** mean MHS per subject.Figure 7.11: Drinking **recovery** mean MHS per subject.

We computed the standard deviation in MHS ratings of the different samples of a subject performing the same activity. The mean standard deviation was 0.56 for brushing teeth, 0.60 for drinking transport, and 0.70 for drinking recovery. This highlights consistency in the score as the rating does not vary a lot between motions performed by the same subject.

We also quantified the separability power of the MHS by optimizing a threshold to maximize the accuracy to predict if the person doing the motion is healthy or has had a stroke. Each motion rated above the threshold is classified as healthy and below as belonging to an impaired participant. The optimal thresholds for brushing teeth, drinking transport, and drinking recovery scores were respectively 9.9, 9.3, and 9.5, leading to accuracies of 0.76, 0.82, and 0.84.

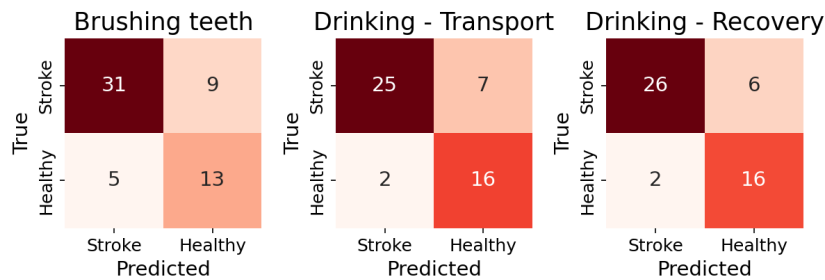


Figure 7.12: Confusion matrices for MHS threshold-based classification of stroke vs. healthy. The MHS threshold are respectively 9.9, 9.3 and 9.5 for brushing teeth, drinking transport, and drinking recovery.

We analyzed the correlation between UEFMA score and brushing teeth, drinking transport, and drinking recovery MHS. The correlation coefficients were 0.37, 0.43, and 0.47, respectively.

7.2 Discussion

We have developed a new data-driven framework called the motion healthiness score (MHS) to evaluate the quality of naturalistic motion. The MHS framework incorporates various metrics to assess the distance between a motion being evaluated and the same motion performed by healthy individuals. The framework has shown good properties like high consistency when rating sample motions from the same subject and good separability with over 80% accuracy between healthy and post-stroke individuals. It provides an opportunity to evaluate a person's impairment in a functional setting without the need for a therapist's evaluation.

This framework offers multiple advantages and can be used to analyze a wide range of motions in both structured and unstructured tasks. A normative dataset is utilized to define what constitutes a healthy motion, without relying on the UEFMA, which does not correlate well with activity capacity. The MHS is an interpretable measure of motion quality and could be used to identify motions that require further training at home. Compared to other impairment measures like UEFMA, the MHS allows for more sensitivity in the moderate and severe range, as there is no maximum distance to healthy data. The MHS provides an opportunity to move impairment measurements into the functional place and could therefore correlate better than other assessments with arm use and independence.

We have identified some areas that require further consideration before using the MHS measure. This measure needs additional validation and future studies should investigate the effect of varying the selected metrics. Although the rating system has desirable properties, its correlation with expert-perceived motion quality should be investigated. Additionally, the scoring system assumes that a stroke participant's motion should approximate that of a healthy participant during recovery. However, the optimal motion solution for a participant with a stroke may differ significantly from that of a healthy participant. One key question to answer is defining the Minimal Clinically Important Difference (MCID) of the distance to healthy data. Using the MHS to track the four to twelve weeks of recovery that take place during outpatient therapy could offer a better understanding of this impairment measure.

One limitation is that the data needs to conform to a multivariate normal distribution. For some activities, there could be different clusters of how healthy people perform an activity. For example, brushing hair and combing hair might produce different measurements of a similar task. In such cases, it may be considered to create more than one ellipsoid for the healthy data, with two or more regions of the PC space being associated with healthy behavior. Including information on the age and gender of the subject can also be beneficial as motions can vary significantly with age, resulting in slower motions, for example.



Table 7.2: Mean Mahalanobis distance to healthy data and MHS for each subject and activity

Subject	Impairment	UEFMA	Brushing teeth		Drinking transport		Drinking recovery	
			D_M	MHS	D_M	MHS	D_M	MHS
C04	healthy	66	3.25	10.00	3.64	9.75	3.38	9.95
C05	healthy	66	2.03	10.00	2.75	10.00	2.08	10.00
C07	healthy	66	1.98	10.00	2.23	10.00	7.35	6.60
C09	healthy	66	5.46	7.11	2.15	10.00	2.05	10.00
C11	healthy	66	2.08	10.00	2.53	10.00	2.43	10.00
C12	healthy	66	2.38	10.00	1.54	10.00	1.79	10.00
C15	healthy	66	2.34	10.00	2.81	10.00	2.42	10.00
C19	healthy	66	4.32	8.83	1.89	10.00	1.97	10.00
C20	healthy	66	2.21	10.00	2.69	10.00	2.07	10.00
C22	healthy	66	2.51	10.00	N/A	N/A	N/A	N/A
C23	healthy	66	6.05	6.21	2.94	10.00	3.82	9.58
C24	healthy	66	2.48	10.00	1.49	10.00	2.13	10.00
C25	healthy	66	2.65	10.00	1.85	10.00	1.40	10.00
C26	healthy	66	N/A	N/A	2.06	10.00	2.26	10.00
C27	healthy	66	2.50	10.00	3.44	9.91	3.83	9.58
C28	healthy	66	4.77	8.16	2.15	10.00	2.16	10.00
C29	healthy	66	2.58	10.00	4.11	9.36	1.81	10.00
C30	healthy	66	2.46	10.00	7.89	6.25	3.61	9.76
C31	healthy	66	7.06	4.69	7.23	6.80	6.81	7.06
s05	mild	65	12.01	0.00	6.89	7.07	7.71	6.30
s08	mild	63	8.15	3.05	N/A	N/A	N/A	N/A
s26	mild	63	7.10	4.63	17.06	0.00	18.36	0.00
s23	mild	61	6.91	4.92	6.39	7.49	6.64	7.20
s16	mild	61	2.90	10.00	3.19	10.00	2.89	10.00
s30	mild	61	11.01	0.00	8.68	5.61	9.03	5.19
s37	mild	61	3.41	10.00	2.23	10.00	2.87	10.00
s10	mild	59	7.93	3.38	N/A	N/A	N/A	N/A
s13	mild	59	3.87	9.52	N/A	N/A	N/A	N/A
s43	mild	58	7.14	4.57	N/A	N/A	N/A	N/A
s32	mild	58	2.37	10.00	3.13	10.00	3.08	10.00
s09	mild	57	9.60	0.85	8.29	5.93	8.33	5.78
s33	mild	57	7.33	4.29	7.96	6.20	8.27	5.82
s42	mild	56	8.47	2.56	7.93	6.23	9.18	5.05
s28	mild	56	7.50	4.03	39.65	0.00	18.56	0.00
s41	mild	55	5.70	6.75	3.62	9.76	2.71	10.00
s35	mild	55	4.48	8.59	3.83	9.59	5.62	8.07
s47	mild	54	2.37	10.00	3.43	9.92	3.30	10.00
s12	mild	53	4.54	8.50	4.68	8.89	8.29	5.80
s36	moderate	51	3.04	10.00	4.23	9.26	3.60	9.77
s44	moderate	49	8.41	2.65	8.85	5.47	9.36	4.91
s17	moderate	48	4.04	9.26	8.20	6.00	8.62	5.53
s46	moderate	48	N/A	N/A	8.74	5.56	7.65	6.35
s03	moderate	48	7.83	3.53	7.85	6.29	8.94	5.26
s27	moderate	47	8.29	2.83	N/A	N/A	N/A	N/A
s11	moderate	47	3.30	10.00	2.88	10.00	8.31	5.79
s53	moderate	45	6.06	6.20	8.15	6.05	9.48	4.80
s54	moderate	45	7.95	3.35	N/A	N/A	N/A	N/A
s06	moderate	42	2.76	10.00	6.46	7.43	8.85	5.34
s01	moderate	42	3.41	10.00	5.76	8.01	5.27	8.36
s19	moderate	38	7.70	3.72	11.53	3.28	12.80	2.00
s39	moderate	37	5.70	6.75	8.19	6.01	5.24	8.39
s20	moderate	37	8.02	3.24	6.96	7.02	8.37	5.74
s07	moderate	34	5.58	6.92	N/A	N/A	N/A	N/A
s45	moderate	34	3.39	10.00	N/A	N/A	N/A	N/A
s22	moderate	34	8.25	2.90	8.11	6.08	9.92	4.43
s02	moderate	32	N/A	N/A	8.47	5.78	11.22	3.34
s24	moderate	32	7.75	3.66	6.64	7.29	7.32	6.63
s18	moderate	31	7.64	3.82	7.42	6.64	8.42	5.70
s04	moderate	27	7.52	3.99	7.40	6.66	8.21	5.88
s25	moderate	26	4.59	8.43	N/A	N/A	N/A	N/A
s21	severe	8	7.89	3.44	N/A	N/A	N/A	N/A

Chapter 8

Conclusion

In conclusion, this research aimed to investigate the ability of inertial measurement units (IMUs) to segment and analyze upper extremity (UE) movements during activities of daily living (ADLs). The capability to do so could help encourage stroke survivors to use their affected arm more throughout the day. We have demonstrated the ability to accurately segment UE ADLs from raw IMU data using a deep learning algorithm that utilizes both CNN and biLSTM layers. While most existing work focuses on classifying a pre-segmented window as part of an activity, we focused on precisely segmenting continuous non-pre-segmented data. The algorithm was tested on ten healthy participants from the Multi-Sensory dataset. The six segmented activities were opening and closing a door, brushing teeth, drinking from a glass, pouring water, and cleaning a table. One of the unique features of our algorithm is the use of context-aware predictions, taking into account past and future data. We achieved respectively F1-scores of 0.83 in across-subject validation and 0.87 in within-subject validation, using only two IMUs located on the forearms. We proposed a methodology for collecting an annotated naturalistic IMU-based ADL dataset in a controlled environment, using video ground truth to reliably label ADLs. We collected data from five stroke patients and five healthy participants and used this data to demonstrate the proof-of-concept of our UE ADL segmentation algorithm on two stroke patients. Our results highlight the potential need for tailored models for post-stroke individuals due to their high motion variability. From the StrokeRehab open-source dataset, we isolated a total of 182 samples of stroke participants brushing their teeth and 148 sample drinking events. In addition, we gathered 86 samples of healthy participants brushing their teeth and 88 samples of healthy participants drinking. Based on the wearable sensor data, we identified robust metrics that can be directly computed and effectively distinguish between healthy and stroke participants' motions during these two ADLs. Finally, we developed a method to aggregate these metrics into a Motion Healthiness Score (MHS). This score uses the Mahalanobis distance to measure the disparity between the movement of a stroke survivor and the same motion executed by healthy subjects. It showed good properties such as high consistency when rating sample motions from the same subject and good separability ($>80\%$ accuracy) between healthy and post-stroke individuals.

We want to acknowledge certain limitations of our study despite the promising results. (1) The first limitation of this research comes from the use of open-source datasets with important information missing. The datasets we used did not include video data, which made it difficult to interpret the results of our algorithms. For example, we were unable to determine why the algorithm made certain false detections for activity segmentation. Additionally, for the motion healthiness score, we could not verify whether motions that were considered close to healthy appeared healthy or not. The StrokeRehab dataset also lacked crucial information, such as the age of healthy subjects. (2) Activity segmentation should still be evaluated in stroke participants with moderate or severely affected stroke participants. (3) Our algorithms were tested in controlled environments and we need to evaluate their performance in a home setting. (4) Our analysis of motion quality left out some important aspects such as compensation, which is a good indicator of bad use. (5) The MHS requires further evaluation. Analysis should be



extended for more tasks, and the minimal clinically important differences (MCID) remain to determine for different tasks.

We also highlight some future steps that should be taken to further advance toward a home-based system for monitoring rehabilitation progress at home. (1) An effort should be made to reduce the cost of data collection or the quantity of data required by deep learning algorithms. Indeed, data collection and labeling through video is a very expensive process. One solution could be the self-labeling of data by the user who records his motion by pressing a button on an app. We can also investigate methods such as self-supervised learning to reduce the reliance on labeled data. (2) We believe it is important to collect data at home to test the performance of the algorithms in an uncontrolled setting. (3) The algorithm for activity segmentation should also be adapted for bi-manual tasks, as stroke participants tend to use both their arms even for typically considered uni-manual tasks like brushing teeth when trying to integrate their affected arm into tasks, as they might lack the ability to use only the impaired arm. More testing on impaired participants is required. (4) Further evaluating the optimal number of wearable IMU sensors to not burden the user and still be able to provide useful information is required. Different configurations can go from only one IMU on the paretic forearm, to multiple sensors, potentially able to provide more information on joint angles and compensation strategies. Acceptability by the end user is critical and this is an important trade-off. (5) The MHS could be validated either by having multiple OTs rating motions and checking the correlation with the MHS, or by following patients' recovery post-stroke.

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Appendix

Appendix A

Evaluation metrics for activity segmentation

Confusion matrix

The confusion matrix is a pivotal tool for assessing the performance of classification models. Represented as a table, the confusion matrix compares the actual classifications of instances with the predictions made by the model. Let C be the confusion matrix, where C_{ij} denotes the count of instances belonging to class i that were predicted as class j . The matrix is organized into rows and columns, where each row corresponds to the instances in a true class, and each column represents the instances predicted by the model for a given class. The diagonal elements (C_{ii}) signify the number of correct predictions for each class, while the nondiagonal elements indicate misclassifications. The confusion matrix can be expressed as follows.

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

This detailed breakdown allows for a nuanced understanding of a model's performance, enabling the computation of various metrics such as accuracy, precision, recall, and F1 score, which are crucial for a comprehensive evaluation of the model's effectiveness in handling multiple classes.

Accuracy

The accuracy of a classifier with multiple labels is computed as follows:

$$\text{Accuracy} = \frac{\sum_{i=1}^n C_{ii}}{\sum_{i=1}^n \sum_{j=1}^n C_{ij}} \quad (\text{A.1})$$

Where:

- Accuracy is the ratio of correctly predicted instances to the total number of instances.
- C is the confusion matrix, where C_{ij} represents the number of instances of class i that were predicted as class j .
- n is the number of classes in the classification problem.



Multi-Class F1 Score

The F1 score is a widely used metric for assessing the performance of a classifier, particularly in scenarios with imbalanced class distribution. In the context of a multi-class classification task, the F1 score for each class can be expressed in terms of the confusion matrix:

$$\text{Precision}(\text{class} = a) = \frac{C_{aa}}{\sum_{i=1}^n C_{ia}} \quad (\text{A.2})$$

$$\text{Recall}(\text{class} = a) = \frac{C_{aa}}{\sum_{i=1}^n C_{ai}} \quad (\text{A.3})$$

$$\text{F1 Score}(\text{class} = a) = \frac{2 \times \text{Precision}(\text{class} = a) \times \text{Recall}(\text{class} = a)}{\text{Precision}(\text{class} = a) + \text{Recall}(\text{class} = a)} \quad (\text{A.4})$$

The F1 scores for each class can then be averaged in different ways. Standard averaging methods include the weighted average, which uses the number of samples as weights; the macro average, the arithmetic mean of the F1 scores; and the micro average, equal to the overall accuracy.

For a multiclass classification task, let $F1(\text{class} = a)$ represent the F1 score for class a as computed in Equation (3). Then, the averaging methods are defined as follows:

1. **Weighted Average:**

$$\text{Weighted F1} = \frac{\sum_{a=1}^n F1(\text{class} = a) \times \text{Number of samples in class } a}{\sum_{a=1}^n \text{Number of samples in class } a} \quad (\text{A.5})$$

2. **Macro Average:**

$$\text{Macro F1} = \frac{1}{n} \sum_{a=1}^n F1(\text{class} = a) \quad (\text{A.6})$$

3. **Micro Average** (equal to overall accuracy):

$$\text{Micro F1} = \frac{\sum_{a=1}^n C_{aa}}{\sum_{i=1}^n \sum_{j=1}^n C_{ij}} \quad (\text{A.7})$$

Comprehensive metrics for continuous activity recognition

In [100], the authors insist that standard evaluation metrics borrowed from the pattern recognition field fail to capture common artifacts in continuous activity recognition, such as event fragmentation, event merging, or timing offsets. Their goal was to characterize performance by showing the algorithm's strengths and weaknesses.

They report true positives and true negatives as well as the following types of error [58]:

- **Overfill and underfill.** Overfill occurs when a start label starts before the actual start time of the activity or when a stop ends after the end of the class. Conversely, underfill happens when the start is after the actual activity starts or the stop is before.
- **Insertion:** Predicting an activity when the ground truth is negative class.
- **Fragmentation:** Error of predicting the negative class between an uninterrupted activity class.
- **Deletion:** Assigning a negative class label where there is an activity.
- **Substitution:** Activity is classified as a different class than the negative one.

Appendix B

IMU-based metrics comparisons



Metric	Healthy		Mild		Moderate		ANOVA		Healthy vs. Mild		Healthy vs. Moderate		Mild vs. Moderate	
	mean	SD	mean	SD	mean	SD	p-value	F-stat	p-value	t-stat	p-value	t-stat	p-value	t-stat
Forearm acc median	75.0	36.5	57.9	26.8	56.2	13.5	<0.0001	11.28	0.005	3.23	0.0001	4.24	1	0.48
Forearm acc 95p	267.5	130.6	187.3	101.4	166.3	65.1	<0.0001	21.41	0.0002	4.17	<0.0001	6.13	0.41	1.50
Forearm acc mead	32.5	17.9	22.4	12.7	20.7	5.8	<0.0001	18.67	0.0003	3.97	<0.0001	5.53	0.93	1.02
Forearm acc SPARC	-5.51	0.97	-6.02	1.29	-7.57	2.07	<0.0001	39.65	0.02	2.73	<0.0001	8.16	<0.0001	5.37
Forearm gyr median	0.473	0.115	0.624	0.212	0.668	0.194	<0.0001	27.27	<0.0001	-5.58	<0.0001	-7.83	0.56	-1.33
Forearm gyr 95p	0.619	0.089	0.731	0.167	0.770	0.162	<0.0001	24.85	<0.0001	-5.31	<0.0001	-7.42	0.46	-1.43
Forearm gyr mead	0.105	0.036	0.071	0.035	0.068	0.038	<0.0001	26.24	<0.0001	5.92	<0.0001	6.45	1	0.55
Forearm gyr SPARC	-5.23	0.69	-5.99	0.89	-6.39	1.06	<0.0001	35.61	<0.0001	5.96	<0.0001	8.29	0.045	2.46
Upper arm acc median	44.8	25.8	48.6	53.4	31.4	7.3	0.0046	5.51	1	-0.57	0.0001	4.42	0.017	2.80
Upper arm acc 95p	125.4	72.4	138.9	121.8	76.33	25.32	<0.0001	12.36	1	-0.85	<0.0001	5.64	0.0001	4.40
Upper arm acc mead	18.87	11.66	20.74	25.33	11.28	3.25	0.0006	7.69	1	-0.60	<0.0001	5.53	0.0043	3.25
Upper arm acc SPARC	-6.14	1.15	-5.87	1.05	-8.01	1.93	<0.0001	50.46	0.40	-1.51	<0.0001	7.57	<0.0001	8.23
Upper arm gyr median	0.245	0.082	0.507	0.235	0.549	0.229	<0.0001	59.51	<0.0001	-9.52	<0.0001	-11.38	0.85	-0.73
Upper arm gyr 95p	0.384	0.096	0.578	0.217	0.626	0.188	<0.0001	44.91	<0.0001	-7.34	<0.0001	-10.39	0.47	-1.43
Upper arm gyr mead	0.061	0.029	0.040	0.030	0.039	0.039	<0.0001	11.02	0.0001	4.36	0.0003	3.99	1	0.13
Upper arm gyr SPARC	-5.81	1.09	-6.89	1.23	-6.78	1.08	<0.0001	22.27	<0.0001	5.80	<0.0001	5.71	1	-0.58
Arm cross corr gyr	0.85	0.36	0.53	0.60	0.49	0.70	0.0001	10.13	0.0002	4.16	0.0001	4.22	1	0.37
Movement time	1.17	0.40	1.16	0.40	1.85	1.20	<0.0001	20.34	1	0.21	<0.0001	-4.89	<0.0001	-4.55

Table B.1: Drinking transport metrics and statistical tests. acc refers to the norm of the acceleration. gyr refers to the norm of the angular velocity. 95p refers to the 95th percentile of the variable of interest and mead is the median absolute deviation. Arm cross corr gyr is the cross-correlation between angular velocities of the forearm and the upper arm.



Metric	Healthy		Mild		Moderate		ANOVA		Healthy vs. Mild		Healthy vs. Moderate		Mild vs. Moderate	
	mean	SD	mean	SD	mean	SD	p-value	F-stat	p-value	t-stat	p-value	t-stat	p-value	t-stat
Forearm acc median	146.7	61.3	106.4	61.0	95.1	42.0	<0.0001	19.28	0.0002	4.06	<0.0001	6.18	0.58	1.30
Forearm acc 95p	284.6	106.2	196.8	101.0	220.3	86.2	<0.0001	16.72	<0.0001	5.20	0.0001	4.20	0.40	-1.51
Forearm acc mead	52.7	21.0	36.9	23.4	37.2	19.5	<0.0001	14.38	0.0001	4.39	<0.0001	4.84	1	-0.08
Forearm acc SPARC	-5.43	1.19	-6.00	1.46	-6.94	1.80	<0.0001	20.60	0.03	2.66	<0.0001	6.32	0.002	3.43
Forearm gyr median	0.370	0.115	0.575	0.220	0.593	0.236	<0.0001	32.59	<0.0001	-7.42	<0.0001	-7.74	1	-0.49
Forearm gyr 95p	0.645	0.083	0.742	0.158	0.783	0.146	<0.0001	23.75	<0.0001	-4.88	<0.0001	-7.47	0.32	-1.62
Forearm gyr mead	0.113	0.058	0.090	0.057	0.074	0.048	0.0001	10.34	0.042	2.48	<0.0001	4.61	0.23	1.78
Forearm gyr SPARC	-4.95	0.86	-5.86	0.97	-6.19	1.14	<0.0001	33.94	<0.0001	6.20	<0.0001	7.85	0.20	1.85
Upper arm acc median	72.9	42.9	68.3	50.6	45.3	15.4	<0.0001	11.28	1	0.62	<0.0001	5.35	0.0006	3.80
Upper arm acc 95p	169.1	101.8	147.6	102.7	102.1	36.2	<0.0001	12.59	0.59	1.30	<0.0001	5.47	0.0011	3.64
Upper arm acc mead	25.9	14.5	24.1	15.9	16.9	6.7	<0.0001	10.75	1	0.78	<0.0001	5.03	0.0012	3.61
Upper arm acc SPARC	-5.45	1.04	-5.95	1.56	-7.02	1.70	<0.0001	24.53	0.06	2.36	<0.0001	7.16	3.96	0.0003
Upper arm gyr SPARC	-5.60	1.08	-6.42	1.06	-6.72	0.97	<0.0001	24.93	<0.0001	4.70	<0.0001	6.87	0.24	1.76
Upper arm gyr median	0.247	0.079	0.512	0.236	0.539	0.252	<0.0001	52.05	<0.0001	-9.64	<0.0001	-10.11	1	-0.68
Upper arm gyr 95p	0.424	0.089	0.605	0.220	0.653	0.182	<0.0001	41.33	<0.0001	-6.91	<0.0001	-10.27	0.46	0.46
Upper arm gyr mead	0.046	0.033	0.051	0.038	0.033	0.033	0.006	5.29	1	-0.95	0.051	2.41	0.008	3.06
Arm cross corr gyr	0.81	0.45	0.63	0.59	0.53	0.7	0.009	4.83	0.09	2.18	0.008	3.07	1	0.91
Movement time	1.00	0.30	1.05	0.32	1.42	0.45	<0.0001	30.92	1	-0.87	<0.0001	-6.95	<0.0001	-5.70

Table B.2: Drinking recovery metrics and statistical tests. acc refers to the norm of the acceleration, gyr refers to the norm of the angular velocity. 95p refers to the 95th percentile of the variable of interest and mead is the median absolute deviation. Arm cross corr gyr is the cross-correlation between angular velocities of the forearm and the upper arm.



Metric	Healthy		Mild		Moderate	ANOVA	Healthy vs. Mild		Healthy vs. Moderate		Mild vs. Moderate	
	mean	SD	mean	SD	mean	SD	p-value	F-stat	p-value	t-stat	p-value	t-stat
Forearm acc median	298.6	149.6	237.5	151.3	160.4	103.6	<0.0001	23.88	0.028	2.63	<0.0001	7.31
Forearm acc 95p	663.4	305.4	505.2	299.8	365.3	227.3	<0.0001	26.13	0.003	3.39	0.001	7.31
Forearm acc mean	118.6	59.2	97.4	65.4	63.9	42.7	<0.0001	22.17	0.09	2.21	<0.0001	7.20
Forearm gyr median	0.448	0.140	0.672	0.243	0.658	0.242	<0.0001	29.25	<0.0001	-7.32	<0.0001	-7.02
Forearm gyr 95p	0.501	0.139	0.699	0.237	0.684	0.234	<0.0001	23.76	<0.0001	-6.61	<0.0001	-6.29
Forearm gyr mean	0.021	0.011	0.012	0.009	0.011	0.009	<0.0001	30.39	<0.0001	5.96	<0.0001	7.01
Upper arm acc median	145.9	80.2	150.2	113.5	90.7	43.3	<0.0001	14.93	1	-0.28	<0.0001	5.87
Upper arm acc 95p	320.8	156.7	339.2	223.6	212.5	106.1	<0.0001	15.56	1	-0.62	<0.0001	5.51
Upper arm acc mean	58.2	30.1	64.0	48.9	36.4	18.3	<0.0001	16.76	1	-0.93	<0.0001	5.99
Upper arm gyr median	0.321	0.132	0.688	0.313	0.651	0.357	<0.0001	42.19	<0.0001	-9.93	<0.0001	-8.04
Upper arm gyr 95p	0.356	0.133	0.705	0.305	0.667	0.347	<0.0001	40.00	<0.0001	-9.67	<0.0001	-7.78
Upper arm gyr mean	0.015	0.007	0.007	0.006	0.007	0.007	<0.0001	36.66	<0.0001	7.29	<0.0001	7.50
Shoulder flexion median	54.04	14.62	55.81	14.07	43.49	18.67	<0.0001	15.77	1	-0.80	0.0001	4.20
Shoulder flexion 95p	58.99	15.38	60.46	15.44	46.90	19.32	<0.0001	17.73	1	-0.61	<0.0001	4.63
Shoulder flexion mean	2.30	1.32	2.16	1.52	1.49	0.95	<0.0001	10.64	1	0.63	<0.0001	4.78
Shoulder abduction median	37.22	19.82	44.81	21.48	30.80	18.38	<0.0001	11.14	0.06	-2.38	0.07	-2.27
Shoulder abduction 95p	44.63	20.60	51.31	20.64	39.94	18.97	<0.0001	13.34	0.11	-2.10	0.01	2.96
Shoulder abduction mean	3.11	2.13	2.74	2.42	2.12	1.36	0.003	5.81	0.87	1.06	0.0006	3.79
Shoulder rotation median	44.13	17.23	38.22	16.02	37.27	19.18	0.021	3.90	0.068	2.30	0.037	2.52
Shoulder rotation 95p	51.97	17.64	44.20	18.48	42.69	20.96	0.003	5.93	0.018	2.79	0.0048	3.21
Shoulder rotation mean	3.48	2.01	2.53	1.28	2.21	1.57	<0.0001	14.16	0.0011	3.65	<0.0001	4.76
											0.45	1.45

Table B.3: Brushing teeth metrics and statistical tests. acc refers to the norm of the acceleration. gyr refers to the norm of the angular velocity. 95p refers to the 95th percentile of the variable of interest and mean is the median absolute deviation.